

REVIEW ARTICLE



Polymer
COMPOSITES

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Review of the past and recent developments in functionalization of graphene derivatives for reinforcement of polypropylene nanocomposites

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Abstract

Graphene has become the promising reinforcing candidate for improvement of polymer composites. However, graphene easily stacks into piles due to the presence of van der Waals forces between graphene layers and make it tremendously hard to be dispersed homogeneously in different polar polymer matrix. For years, some dispersion methods, such as chemical functionalization, melt mixing, solution mixing, in situ polymerization, to enhance the dispersion of nanofillers in polymer matrix have been demonstrated. Therefore, several functionalization methods of graphene, synthesis of composites approaches, characterizations techniques of graphene-polypropylene composites are highlighted in this comprehensive review. This review also reveals that the addition of very

Abbreviations: ACNT, alkylated carbon nanotube; AGO, alkylated graphene oxide; ARGO, alkylated reduced graphene oxide; CB, carbon black; CC, cyanuric chloride; CNT, carbon nanotube; CO₂, carbon dioxide; COOH, carboxyl group; CPE, chlorinated polyethylene; DI, distilled water; DMAE, 2-(2-[dimethylamine]-ethoxy) ethanol; DMF, N,N-dimethylformamide; DTG, derivative thermogravimetry; EDS, energy-dispersive spectroscopy X-ray; EG, expanded graphite; ESEM, environmental scanning electron microscope; FESEM, field emission scanning electron microscope; FG, functionalized graphene sheets; FTIR, Fourier transform infrared; GIC, graphite intercalated compounds; GN, graphene nanosheets; GNP, graphene nanoplatelets; GNS, graphene nanosheets; GO, graphene oxide; GONR, graphene oxide nanoribbons; H₂S, hydrogen sulphide; H₂SO₄, sulphuric acid; H₃PO₄, phosphoric acid; HA-GO, hexylamine-graphene oxide; GO-HD, alkylated graphene oxide; HDPE, high-density polyethylene; HMDA, hexamethylene diamine; HNO₃, nitric acid; IS, in situ; KN1, kneading block 1; KN2, kneading blocks 2; KN3, kneading blocks 3; LLDPE, low density polyethylene; MA, maleic anhydride; MAO, methylaluminoxane; MAPP, maleic anhydride grafted polypropylene; MC, melt compounded; MDI, 4,4'-diphenylmethane diisocyanate; MOF, metal-organic frameworks; MWCNT, multi-walled carbon nanotube; N₂O₄, dinitrogen tetroxide; β-NA, β-form nucleating agent; NCO, isocyanate group; NH₂-PEG-NH₂, bis(3-aminopropyl)-terminated poly(ethylene glycol); NO₂, nitrogen dioxide; NO₃, nitrate; OBCs, olefin block copolymers; ODA, octadecylamine; OH, hydroxyl; OP-10, polyoxyethyleneoctylphenyl-10; OTES, octyltriethoxysilane; PA, pimelic acid; PAA, polyacrylic acid; PAN, polyacrylonitrile; PC, polycarbonate; PE, polyethylene; PET, polyethylene terephthalate; PEN, poly(ethylene-2,6-naphthalate); PI, polyimides; PLA, polylactide; PMMA, poly(methyl methacrylate); PP, polypropylene; PPD, P-phenylenediamine; PPG, polypropylene graphene; PPGO, polypropylene composites based on pristine GO; PP-OH, hydroxylated PP; PPMA, maleic anhydride-grafted PP; PS, polystyrene; RF, ramie fiber; RGO, reduced graphene oxide; SEM, scanning electron microscopy; SO₂, sulphur dioxide; TDT, thermal degradation temperature; TEA, triethylaluminum; TEM, transmission electron microscopy; TGA, thermogravimetric Analysis; TMB-5, aryl amides compound; TOAB, tetra-n-octylammonium bromide; UV-vis, ultraviolet visible; WAXD, wide angle X-ray diffraction; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction.

low amount of nanofiller able to enhance the mechanical properties of composites considerably. With this review, a clear direction is beneficial for the researchers to solve the dispersion issues and to enhance interfacial adhesion to produce better mechanical properties of composites. The application of nanocomposite, an outlook of the challenges and different approaches that can practically useful to tackle important issues are also presented.

KEYWORDS

electron microscopy, graphene, mechanical properties, nanocomposites, polypropylene

1 | INTRODUCTION

Carbon nanomaterials have been widely used as nanofillers in the field of composites. Two important criteria that affect the synthesis of nanocomposite are the properties and degree dispersion of nanofillers in polymer matrix. The discovery of numerous carbon nanofillers, such as carbon nanotubes (CNTs), fullerene, carbon black, carbon nanofibers, and graphene significantly enhances the properties of neat polymers. Thus, the number of publication about carbon nanofillers-based nanocomposites is increasing from year to year. Recently, the mechanical and electrical properties of nanocomposites are two most concerned features to focus by scientists to evaluate the performances of the polymeric nanocomposites. Among several nanofillers, many researchers pay very much attention on graphene because of its excellent properties in various applications. Graphene is a mono layer carbon in the crystalline honeycomb graphite lattice with sp^2 hybridized carbon atoms, strongly bonded in hexagonal rings which make it less favorable to be applied as nanofiller when mixing with polar polymer.^[1]

On the other hand, graphene oxide (GO) and graphite oxide are modified graphitic forms, along with expanded graphite and graphite intercalated compounds. Graphite oxide consist of multilayer structure whereas GO mostly appears as mono-, bi- or few layers graphene sheets. GO offers much more potentials than graphite oxide in satisfying needs for wider applications because of few layers structure, which reduces the agglomeration issue in mixing process with substrate.^[1] Unlike graphite and pristine graphene, GO contains abundant reactive oxygen-containing functional groups which allows further functionalization at its structure for various applications in last decade, such as cancer treatment,^[2–5] drug delivery in human body,^[6–11] battery,^[12] aerospace,^[13,14] and others. Therefore, graphene derivatives can be incorporated with varieties of functional materials, such as inorganic nanostructures, organic crystals, polymers,

metal-organic frameworks, biomaterials, and CNTs.^[15] Moreover, some scientists conducted the theoretical studies to improve and analyze functionalization.^[16,17]

Graphene-based polymer nanocomposite has been developed for many years; however, obstacles are still remain that restrict these polymer nanocomposite to show its full potential.^[18] Non-functionalized carbon materials and organic polymers are not compatible with each other due to the presence of van der Waals forces to induce aggregates in carbon materials thus, producing homogenous dispersion in organic polymers are hardly to be achieved. On the contrary, GO comprises many oxygen-containing functional groups, such as hydroxyl, epoxide, diols, ketone, and carboxyls, which make GO compatible with polar polymers. Besides, the presence of carbonyl and carboxyl on the edges of graphene sheets also make GO behaves hydrophilic where GO can be dispersed well in water. As most of the non-polar or organic polymers can be dissolved only in organic solvents instead of aqueous media, homogenous mixing of GO and organic polymers cannot be achieved. Furthermore, GO is not suitable to be used as nanofiller in producing conducting nanocomposites as GO is electrically insulating. Since the aspect ratio of GO is reduced when aggregation happens, the mechanical properties of nanocomposites can be significantly weakened. Therefore, surface modification on GO is needed so that molecular level dispersion for modified GO in non-polar polymer matrix, such as polypropylene (PP), polyethylene, and polystyrene can be attained.^[18] Currently, some effective approaches have been reported which can be categorized as covalent and non-covalent functionalization. Covalent functionalization is preferred to enhance mechanical properties of graphene-based polymer nanocomposites due to the presence of hydroxyl, epoxy, and carboxylic groups as a base for attachment of various types of alkyl chains to facilitate the strong interfacial interaction between the nanofillers and polymer matrix.

The field of graphene-based polymer nanocomposite is dynamic which is driven by creative and revolutionary invention from carbon nanotechnology progress.

Therefore, it inspires the publication of this review to ease researchers to study the past and recent progress as well as to boost state-of-the-art research in this field. In this review paper, it aims to highlight the detailed compilation of the chemical modifications of graphene: GO, reduced graphene oxide (RGO), alkylated graphene oxide (AGO), and alkylated reduced graphene oxide (ARGO) as nanofillers and their nanocomposites (Figure 1). In addition, we also incorporated the past and recent characterization methods of functionalized graphene-based nanocomposites for researchers to read quickly and gain benefits. Finally, we present a short summary and an outlook on the major challenges and opportunities for future research.

2 | NANOFILLER PREPARATION METHODS

2.1 | Synthesis of GO

Many research works reported three general methods to synthesize GO, namely Staudenmaier's, Hummers', and Hofmann's Method where different oxidation agents were used.^[19] GO was first demonstrated by Brodie in 1860 where graphite with potassium chlorate (KClO_3) and nitric acid (HNO_3) were mixed. However, this method was regarded as not economical for GO production as it was time consuming and hazardous chemicals were used. For Staudenmaier's method, apart from concentrated sulphuric acid (H_2SO_4), fuming HNO_3 and KClO_3 were also used.^[20] In addition, Hofmann's method used similar chemicals as used in Staudenmaier's method except the

fuming HNO_3 was substituted by the concentrated HNO_3 . On the other hand, HNO_3 as used by Brodie was replaced by sodium nitrate (NaNO_3) and KMnO_4 in Hummers' method.^[21]

Among all GO synthesis approaches, Hummers' method is the most popular in producing GO until now.^[21,22] The reasons are simple as this method allows high yield production and safer in experiment because KMnO_4 was used instead of KClO_3 to avoid the spontaneous explosion during oxidation. Moreover, HNO_3 is also replaced by NaNO_3 to prevent the formation of fog acid.^[23,24] This method was modified again to simplified Hummers' method, modified Hummers' method, and improved Hummers' method due to high interest of most researchers to further improve the quality and amount of GO. Table 1 compiles three different modified methods with reference to the conventional Hummers' method.

2.2 | Synthesis of RGO

Graphene, GO, and RGO have their own respective excellent properties and limitations. Graphene is known for having great mechanical, thermal, and optical properties. However, contrary to graphene, GO is hydrophilic, which enables GO to form stable aqueous colloids thus, expanding the use of GO in many applications. The oxidation can be completed by using many available chemical methods where graphite is used as starting material. The process is very economic and able to produce high yield of GO. On the other hand, reduction of GO to produce RGO allow restoration structure and properties of graphene partially. This can be achieved by using agent

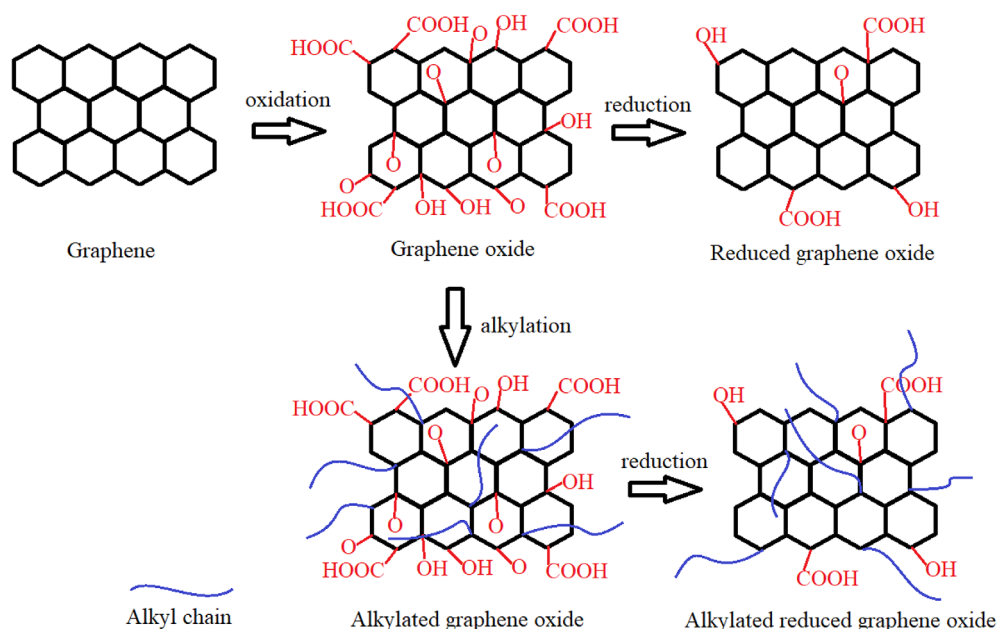


FIGURE 1 Four types of graphene derivatives have been used for the reinforcement of polypropylene nanocomposites [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Comparisons between conventional, simplified, modified and improved Hummers' method

Method	Hummers'	Simplified	Modified	Improved
Chemicals used	Concentrated H ₂ SO ₄ , NaNO ₃ , KMnO ₄	KMnO ₄ , H ₂ SO ₄	Concentrated H ₂ SO ₄ , NaNO ₃ , KMnO ₄	H ₂ SO ₄ , H ₃ PO ₄ , KMnO ₄
Modification step from conventional Hummer's method	Reaction temperature maintained below 20°C during KMnO ₄ addition, then raised to 35°C. After adding DI water, the temperature is maintained at 98°C	NaNO ₃ is eliminated	The amount of NaNO ₃ is reduced but increased the amount of KMnO ₄	H ₂ SO ₄ /H ₃ PO ₄ mixture at 9:1 ratio The use of NaNO ₃ is eliminated Reaction done at room temperature The amount of KMnO ₄ used is doubled than the conventional method
Disadvantages	Release of toxic gasses, such as NO ₂ and N ₂ O ₄ , during oxidation	There is possibility for explosion during KMnO ₄ addition	Release of toxic gasses, such as NO ₂ and N ₂ O ₄ , during oxidation Residual, such as Na ⁺ and NO ₃ ions, is difficult to be removed from the waste water which formed during the processes of synthesizing and purifying GO	Not applicable
Advantages	Not applicable	No toxic gaseous are emitted	Not applicable	No toxic gaseous are emitted Simpler experimental procedure Greater amount of hydrophilic oxidized graphene material to be produced
References	[22,168]	[43,169]	[24]	[22–24,168,170]

reduction to remove the oxygen-containing functional groups which attached at the surface of graphene.^[25]

Reduction can be achieved mainly by two methods: chemical and thermal reduction in which chemical reduction is more preferable especially for large-scale production.^[26] This is due to the fact that, chemical reduction can be done at room temperature and not more than 200°C.^[25] On the contrary, thermal reduction needs to be held at a very high temperature up to 1050°C.^[27–30] Thus, the thermal reduction cost is higher than the chemical reduction.

The conventional reduction methods of GO to RGO was conducted by using varieties of reducing chemical reagents. Hydrogen sulphide (H₂S) was the first reducing agent used in 1934^[31] and followed by varieties of other chemicals, such as hydrazine and hydroquinone.^[32,33]

Hydrazine is the most commonly used and proved to be the most effective of agent reduction.^[26] It is because the RGO produced by using hydrazine showed improved electrical and structural properties, almost similar with pristine graphene.^[34] However, hydrazine is very toxic and explosive.^[26,35]

Chen et al demonstrated the use of series of sulfur-containing compounds as reducing agents, such as sodium bisulfite (NaHSO₃), sodium sulfite (Na₂SO₃), sodium thiosulfate (Na₂S₂O₃), sodium sulfide non-hydrate (Na₂S•9H₂O), thionyl chloride (SOCl₂), and sulphur dioxide (SO₂)^[36] where the sulfur-containing compounds, NaHSO₃ and SOCl₂ displayed similar reducing ability as hydrazine. Furthermore, NaHSO₃-reduced graphene showed higher electricity conductivity of 6500 Sm^{−1}, while 5100 Sm^{−1} was measured for

hydrazine-reduced graphene paper. In addition, they also mentioned the benefit of using NaHSO_3 is safer than hydrazine as it is less toxic and non-volatile.^[36]

Wang et al reported the use of hydroquinone as reducing agent. The work started with the production of GO by using Hummers' and Offeman's method with some modifications. The produced GO was exfoliated first before the reduction by refluxing GO using hydroquinone for 20 h.^[32] In addition, Bourlinos et al also reported the use of hydroquinone with sodium borohydride (NaBH_4) as reducing agents. In their work, the comparison was made in order to investigate the performance between NaBH_4 -reduced GO and hydroquinone-reduced GO.^[33] Amazingly, the basal spacing (d_{002}) of both produced crystalline RGO were found similar with d_{002} of 3.39 Å. However, an additional step was required to synthesize crystalline RGO using NaBH_4 where RGO was further calcined in an inert atmosphere. On the contrary, the crystalline RGO prepared using hydroquinone was dried directly under soft thermal conditions.^[33] The additional step increased the cost production. Therefore, hydroquinone is more preferable than NaBH_4 as reducing agent.

Apart from various chemicals, some articles also reported the application of natural reducing agents as an alternative for hydrazine, such as organic acids (such as ascorbic acid), plant extracts (that includes mushroom extracts,^[37] green tea^[38] and carrot juice^[39]) microorganisms (like *Escherichia coli*^[40]), sugars (such as fructose, glucose and sucrose^[41]), antioxidants (such as tannic acid, amino acids, proteins and others), due to their non-toxic and environmental friendly characteristics.^[26] However, some natural reducing agents have the disadvantages because second agent is needed to complete the reduction. Besides, in some cases, stabilizer is added to avoid the aggregation of RGO sheets. Moreover, extra steps might be required to remove any excess reductant or the by-products.^[26]

The first natural reducing agent reported was ascorbic acid in 2010.^[42] Ascorbic acid was chosen, as it is non-toxic and inexpensive.^[26] Zhang et al used the L-ascorbic acid (L-AA) as reducing agents for RGO. The reduction process was conducted in water at room temperature with vigorous stirring and monitored by using UV-Vis spectroscopy. From the absorption spectrum, absorption peaks were first noticed at 230 and 300 nm, proving the presence of GO. However, the intensities for both peaks decreased slowly with increasing reaction times and completely disappeared after 24 h. Interestingly, a new peak can be observed at 264 nm, which proved the presence of RGO.^[42] On the other hand, Chong et al compared the GO reduction efficiency between hydrazine and natural reducing agents, such as lemon juice and

vinegar. They noticed that lemon juice able to reduce GO but hydrazine showed higher efficiency in the GO reduction.^[43] Besides, phosphorous acid also demonstrated its efficiency to reduce GO to RGO in an experiment reported by Long et al.^[44] The reduction was conducted in more simple way by stirring the mixture of GO and phosphorous acid for 24 h at 90°C.

3 | COMPOSITES PREPARATION METHODS

3.1 | Synthesis of graphene-based PP composites using non-alkylation techniques

Polymer processing can be defined as an engineering activity involves transformation of raw polymer materials into a product with desire properties.^[45] The process involves shaping, compounding, and chemical reactions, which lead to macromolecular modifications and morphology stabilizations.^[45] Melt mixing, solvent mixing, and in situ polymerization are the most popular preparation methods of polymer matrix composites.

3.1.1 | Melt mixing

In industry, melt mixing is widely used especially when it involves thermoplastic polymer. It is a technique to disperse nanofillers into polymer matrix under high temperature and high shear forces. This method is actually a mixing process between polymer matrix and filler, without the presence of any solvent. No solvent means no additional processing step needed to remove the solvent; thus, making melt mixing is a very economical approach.

However, when compared with solvent mixing method, melt mixing is generally less effective in filler dispersion in polymers at high loading due to the high viscosities of composites after mixing.^[46] To date, graphene-reinforced composites, such as exfoliated graphite-poly(methyl methacrylate) (PMMA), graphene-PP, GO-poly (ethylene-2,6-naphthalate), and graphene-polycarbonate, have been produced using melt mixing, even with the possibility of poor dispersion of the fillers for many applications.^[47] Therefore, no constraint in the processing polymers via melt mixing, where it is applicable for both polar and non-polar polymers. Moreover, melt mixing is a common technique applied at industries for large-scale production of thermoplastic composite.^[47]

El Achaby et al reported on the mechanical, thermal, and rheological performance of the composites prepared by melt mixing. Interestingly, the composites were

produced by simply mixing graphene nanosheets (GNs) with PP, with the use of only a twin-screw extruder. The GNs were synthesized in the chemical reduction of GO. The performance of the produced composites was compared with neat PP. The group managed to prove the significant enhancement in mechanical and thermal properties.^[48] Therefore, the addition of GNs as reinforcing filler for PP matrix is an advantage.

Zhong et al highlighted the benefit of installation of ultrasonic device at the extruder before the die section to facilitate the dispersion of the filler, such as carbon black, CNT, and graphene nanoplatelets (GNP) in polymer matrix. It is because the vibration from ultrasonic help in breaking the formed agglomerates during melt mixing; thus, improving the dispersion of filler. Figure 2(A) shows the schematic diagram for both twin screw extruder and screw configuration. The screw contains three kneading blocks, denoted as KN1, KN2, and KN3.^[49] Figure 2(B),(C) illustrates the bubble cavitation mechanism to explain the manner of ultrasonic in improving the dispersion of filler in the matrix. These bubbles occurred due to the presence of air that might be trapped inside the melted polymer during extrusion. With the use of ultrasonic, these bubbles can break the agglomerates depends on the location of the bubbles.^[48] As illustrated in Figure 2(B), outside bubbles can cause

the particles to breakaway from the primary agglomerates, whereas as seen in Figure 2(C), the bubbles can break the agglomerates from inside. In other words, both mechanisms could diminish any agglomerate that present when the extrusion is conducted.^[49]

Since melt mixing always causes the re-staking and poor dispersion of graphene sheets in polymer matrix, modification of PP to form terminally hydroxylated PP (PP—OH) using both triethylaluminum and rare-Ethylenebis(1-indenyl) zirconium dichloride (EBIZrCl₂) as a chain transfer agent and catalyst, respectively, was demonstrated.^[50] The researchers named the method as chain transfer method. The PP—OH and GO were heated in tetradecane at 200°C to prepare PP-GO prior to the melt mixing with PP to synthesize PP/PP-GO nanocomposites. To prepare high quality composites, the polymer mixer machine has important effect on the final product. Shear flow field is the main operational design in the traditional polymer mixer equipment which show some disadvantages as compared with elongational flow, such as poor dispersion, and high mixing temperature is required. Vane extruder was obviously helpful in the dispersion and exfoliation of functionalized GO sheets under the elongation flow as reported by Qiu et al.^[51] The mechanical properties of composites were enhanced significantly by vane mixer than previous mixing method by using twin-screw extruder.

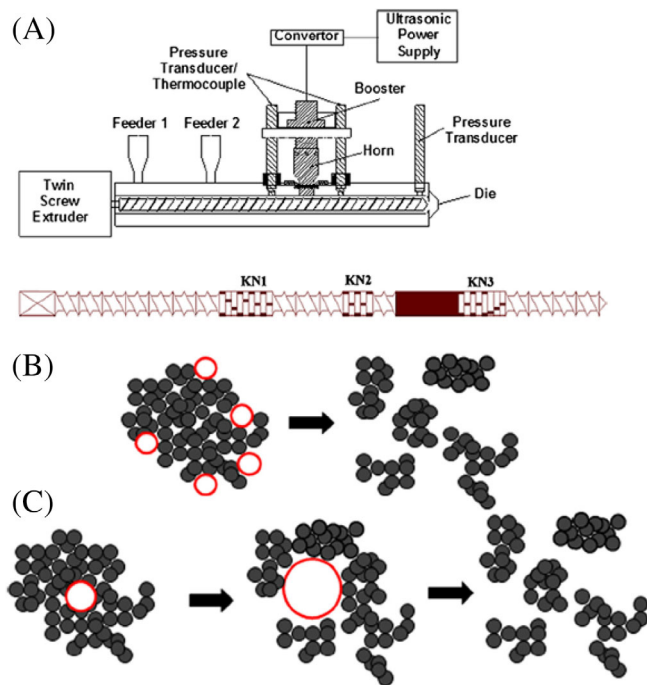


FIGURE 2 (A) Schematic diagram for twin screw extruder and screw configuration (B) Illustration of (B) outside and (C) inside bubble cavitation mechanism. Reproduced with permission from Reference [49]. Copyright 2016, Elsevier [Color figure can be viewed at wileyonlinelibrary.com]

3.1.2 | Solvent mixing

In general, solvent mixing is a method to mix colloidal suspensions of GO platelets or other graphene-based materials with any type of polymer. If the polymer is appeared in solution form, the mixing can be handled easily. However, for the powder form of polymer, it is suggested to be dissolved in the suspension of the filler by stirring or shear mixing. The composites can be collected after the removal of the solution for further testing or application.^[52]

Removal of the solution can be completed either by using precipitation or by casting the suspension into a mold. The precipitation of colloidal suspensions of G-O platelets with desired polymer can be done by adding a non-solvent to allow polymer chains to capture the filler. On the other hand, mold casting was conducted by directly adding the composite suspensions into a mold. However, mold casting is not recommended, as this casting process actually causes the aggregation of filler in the composite especially during solvent removal, which later diminish the good properties of the produced composites.^[19]

Although filler aggregation and stack up can be described as the limitations for solvent mixing, solvent

mixing remains as a simple method in dispersing graphene-based fillers into a polymer matrix. Therefore, in order to form stable filler suspension and also to prevent any filler aggregation and stack up, chemical functionalization and grafting-to/from approaches have been applied. For example, lyophilization methods,^[53] phase transfer techniques,^[54] and surfactants^[55] were employed to facilitate solution mixing of graphene-based composites. In the case of synthesis of nanomaterial PP composites, common solvents that frequently used are water, acetone, N,N-dimethylformamide (DMF), chloroform, DCM, and toluene.^[47] The compatibility of solvents with polar or non-polar polymer is crucial for homogenous mixing. Polar polymers, such as PMMA, polyacrylic acid, polyacrylonitril, and polyesters can be easily dissolved in the solvent before GO was added.^[47] On the other hand, PP is insoluble in most organic solvents even at room temperature due to the fact that PP is a non-polar polymer and has a great chemicals resistance in most conditions.^[56,57]

As an example of solvent mixing, the production of isotactic polypropylene (iPP)/graphene nanosheets (GNS) composites was demonstrated by Xu et al. The purpose of solvent mixing was to prepare nanocomposite for investigation about the associated effect of GNS and shear flow on the crystallization output of iPP. The effective of solvent mixing can be proved with the presence of GNSs where the effects of shear-activated nucleation and direction of iPP crystals were improved.^[58] For comparison with melt mixing, solvent mixing shows that produced composites possess better mechanical and thermal properties. For example, Yang et al reported the production of PP/RGO nanocomposites with improvement in the thermal stability and mechanical properties with every increment of the RGO loadings.^[59]

Furthermore, the combination of two methods able to deliver unexpected outcome as demonstrated by Chaudhry and Mittal.^[60] They found the enhancement of tensile modulus and peak stress in sample which prepared with combination of solution blending between compatibilizer and GO, then followed by melt mixing with high-density polyethylene (HDPE) as compared with similar composite which synthesized by melt mixing only. Compatibilizer that used in this work was chlorinated polyethylene (CPE) with 25% and 35% chlorine content which enables the homogenous dispersion and exfoliation of nanofillers due to the nature of PE has lack polar groups on its backbone. Thus, the low amount of polar groups in PE lead to its poor dispersion in polymer which need compatibilizer to solve the issue. Beside, A. Rosehr et al^[61] also reported two methods to prepare composites where good nanofiller distribution and enhanced mechanical properties of composites were achieved successfully.

There are few works regarding the solvent mixing were reported such as the dispersion of clay in low density polyethylene (LLDPE) by using ethylene acid copolymer as compatibilizer and the thermal properties of nanocomposite were improved successfully.^[62,63] In addition, in the work showed by Kim et al.,^[90] maleic anhydride was used as compatibilizer to mix both of LLDPE and exfoliated graphite nanoplatelets to form composites, whereas You et al^[64] demonstrated the mixing between tryptophan-functionalized graphene and maleic anhydride grafted PP (MAPP) in xylene solution. The authors highlighted that the strong adsorption between graphene and MAPP due to the aid of tryptophan which grafted on graphene.

3.1.3 | In situ polymerization

One thing that differentiates this method from others is that the monomer, propylene is the starting material. In the polymerization reaction, propylene monomers are added sequentially in order to form a long and linear polymer chain as shown in Figure 3(A).^[65] The reaction begins with the dispersion of filler in monomer first

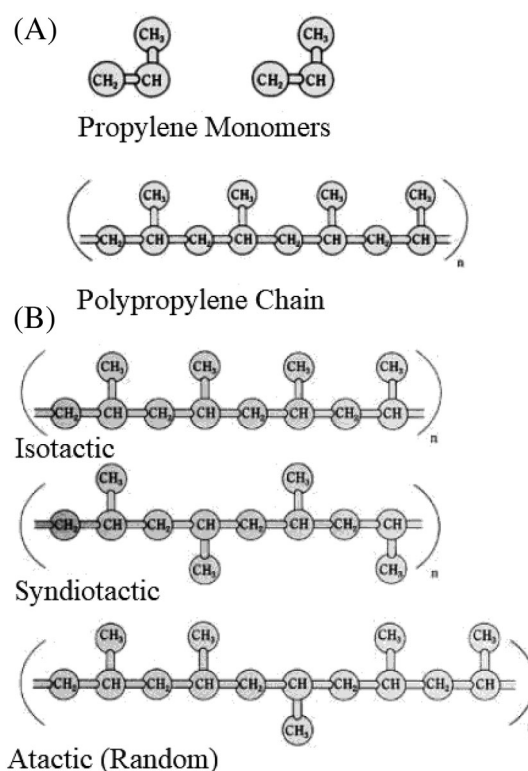


FIGURE 3 (A) Molecules of propylene monomer and polypropylene polymer (B) Chains of different stereoisomers of PP. Reproduced with permission from Reference [65]. Copyright 1998, Elsevier

before proceed with polymerization.^[47] However, this method requires the use of catalyst to facilitate the polymerization. In the PP polymerization, catalyst delivers active or polymerization spots for polymer chain and propylene monomer to react instantly. By definition, catalyst is a material that speeds up the reaction rate but does not have permanent chemical changes.^[65] Ziegler-Natta catalyst is the most common commercial catalyst which was developed by Nobel Prize receiver, Karl Ziegler and Giulio Natta in 1963 for the development of polyolefin polymerization catalysts with high yield and a high degree of stereospecificity^[66] where iPP was produced.^[56,66] This stereospecificity allows the reactants to control the outcome of the reaction and in this case, it is isotactic PP. Other forms of PP including syndiotactic PP and atactic PP are shown in Figure 3(B). Each of these stereoisomers has its unique melting and mechanical properties.^[65]

Besides, Huang et al demonstrated the in situ PP polymerization with GO-supported Ziegler-Natta catalyst by incorporating Mg/Ti species catalyst at the GO individual sheet surface. The polymerization was conducted using a Parr stainless steel autoclave reactor equipped with a mechanical stirrer. Besides Mg/Ti species,^[67] metallocene complex $\text{rac-Me}_2\text{Si(Ind)}_2\text{ZrCl}_2$ and methylaluminoxane (MAO)^[68] have been used as catalyst and as cocatalyst, respectively. Grignard reagents was demonstrated by Huang et al where it has the ability to reduce graphite oxide to graphene sheets and functionalized graphene sheets with few Mg-Cl species in slurry condition in a Parr stainless steel autoclave reactor.^[69] This work covered a very interesting topic where the polymer is insulative material, was transformed into conductive polymers with the addition of conductive fillers. Therefore, the applications for polymers can be expanded into electromagnetic shielding, antistatic, conductive paints, and others.^[69] This method also allows homogenous dispersion of delaminated GO sheets in the PP matrix, indicating that incompatibility between polar nanofiller and non-polar PP matrix can be overwhelmed.^[69]

On the other hand, Milani et al^[68] demonstrated in situ polymerization using the metallocene complex, such as $\text{rac-Me}_2\text{Si(Ind)}_2\text{ZrCl}_2$ as catalyst, whereas MAO was used as a co-catalyst. When compared with neat PP, the produced nanocomposites possessed almost similar molecular weight with polydispersity and tacticity. Besides, Cardoso et al proposed a new approach in producing PP/clay composites. The new approach including the preparation of masterbatch via in situ polymerization and followed by melt mixing.^[70] The proposed method was started with the PP polymerization in a Büchi reactor by using hexane as polymerization solvent. The product

produced from this polymerization was called as masterbatch of PP/clay and then it was melt mixed with polymer in a mini twin-crew extruder.^[70] For comparison between neat polymer and PP/clay composites, they noticed that the storage modulus (E') of composite was enhanced as compared to neat polymer. This proved that the in situ polymerization showed a benefit to disperse clay in the PP matrix where strong interface interaction between filler and matrix was created.^[70]

Furthermore, Cromer et al demonstrated the synthesis iPP-exfoliated graphene nanoplatelet (iPP-xGnP) by using melt compounded (MC-) and in situ (IS-) polymerization.^[71] The composites showed increment in crystallization temperature, mechanical strength, and modulus as compared to pure PP. In addition, the composites displayed increased of Young's modulus and yield stress with increasing xGnP loading. However, the composite that produced by using IS- showed better ductility and toughness than MC-.^[71] Unfortunately, both MC- and IS-produced composites also had the issue of filler agglomeration. Agglomerates with diameter of 1–10 μm can be seen on the surface of both composites under TEM. However, larger agglomerates with diameter greater than 10 μm can be observed in IS- produced composite only. This is likely due to the poor interaction between GnP with toluene which induced flocculation and sedimentation of nanoplatelets.^[72] Therefore, the big agglomerates prohibited the mixing flow field during IS- polymerization due to the lack of mixing shear stress in a low viscosity polymerization solvent. However, smaller agglomerates were formed in MC- method because of the relatively higher mixing viscosity and shear stress.^[71]

3.1.4 | By in situ reduction of GO to RGO techniques

Since the RGO favored to restack to form thick pile structure, therefore, the melt-blending method could not blend well the mixture of RGO and PP. Homogenous mixing of the filler and matrix is very important to produce composites with better mechanical properties. An alternative solution by using PP latex has gained more interest from researchers recently. PP latex is a water-based emulsion of maleic anhydride-grafted-isotactic polypropylene (MA-g-iPP), which is composed of 30 wt% (MA-g-iPP), 5 wt% of anionic surfactant of the oleic acid type, neutralized by an amino alcohol.^[73,74]

Song et al conducted the research by coating the graphene using PP latex before melt-blending with PP matrix to achieve well dispersion of nanofiller.^[73] Figure 4 illustrates the process flow for the synthesis of PP/graphene (PPG). The reaction started with oxidation

of graphite to graphite oxide (GO). The GOs were dispersed in distilled water and PP latex, before adding hydrazine hydrate to produce graphene-coated PP latex. The enhancement in tensile values was achieved with small loading of graphene. In addition, the glass transition temperature and thermal oxidative stability of PP were improved as well.^[73] Similar work was reported by Wang and co-workers using PP latex.^[74] The synthesis of RGO/PP composites was almost similar with Song et al.^[73] except no further melt mixing with PP matrix was carried out.

3.2 | Characterization of graphene-based PP composites using non-alkylation techniques

3.2.1 | SEM/FESEM/TEM

Scanning electron microscopy (SEM) has been used to investigate morphologies for composites.^[75] From the SEM micrographs, the interaction between filler and polymer matrix as well as the presence of agglomerates/aggregates can be observed. Figure 5 shows the SEM micrographs of the composites with loading of 1, 1.5, and 2 wt% of GNs produced by mixing GNs and PP using a twin screw extruder. Large aggregates can be observed on the surface of 1.5 wt% GNs as compared with 1 and 2 wt% loadings. In spite of aggregates were observed, homogeneous distribution of GNs in few areas can be noticed as well which indicated the interaction between GNs and polymer matrix was good to some extent.^[48] Furthermore, the dispersion of nanofillers can be examined using SEM micrographs. Figure 6(A) shows the structure of GNPs with an average diameter of ca. 15 μm . Smooth surface of PP can be seen in Figure 6(B) but the

formation of rougher surface was observed after GNPs were added (Figure 6(C)).^[76] The formation of rough surface could implied an increase energy occurred at the fracture composite surface.^[78] In addition, no aggregations can be observed as the GNPs loading was 5.2 vol%, supporting the findings by Kalaitzidou et al where poor dispersions of filler in PP matrix occurred at 10 vol% of filler loading.^[77] Besides, the size of nanofiller has some significant effects to the composite. The agglomeration size of xGNP-15 as big as 200–300 μm was seen in Figure 7(A) as indicated by the arrows. However, Figure 7(B) shows that xGNP-1 was better dispersed in PP with smaller and lesser agglomerates due to the stiffer and smaller size of xGNP-1 as compared with xGNP-15 with bigger size (diameter of 15 μm and aspect ratio of ~ 1500) and more flexible structure. Thus, the morphology of xGNP-15 was gone to remain its effectiveness as reinforcement nanofiller in the xGNP-15 composites.^[78]

The fracture surface revealed better understanding about the GO nanosheets dispersion under inspection by using SEM. The smooth surface of neat PP (Figure 8(B)) was transformed to rough surface with the presence of GO whereas GO nanosheets were found well dispersed in the matrix (Figure 8(C),(E)). Furthermore, with the increase of GO content, the fracture surface showed the appearance of GO sheet significantly. In Figure 8(E), the composite fracture surface was filled with GO nanosheets completely at 1.52 wt% due to GO sheets have large surface.^[67] Besides SEM, transmission electron microscopy (TEM) provides better observation to differentiate graphene and PP at the cross section of sample due to the different brightness of graphene (dark colour) against PP can be noticed as shown in Figure 9(A),(C), (D). The surface of reduced graphene was coated with PP latex as marked with red circles (Figure 9(B)). After reduction, almost all oxygen-containing functional

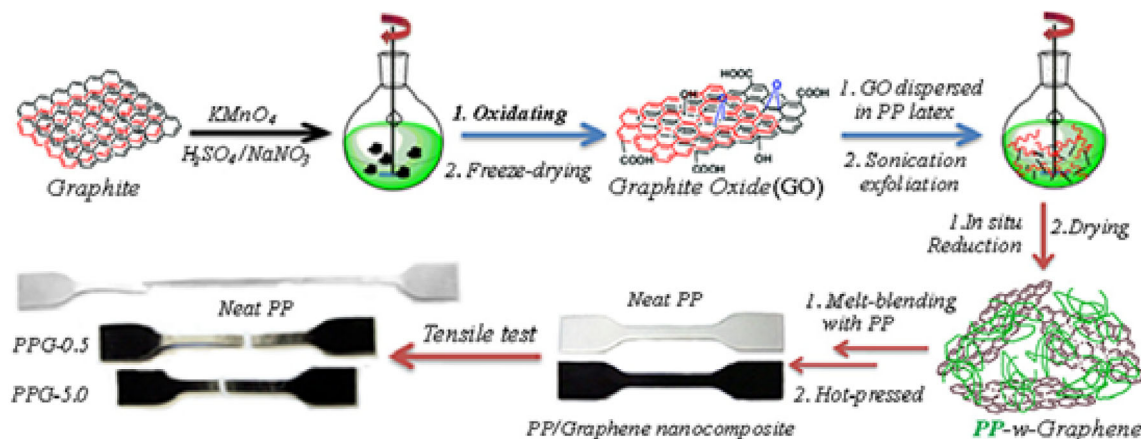


FIGURE 4 Production of polypropylene (PP)/graphene (PPG) nanocomposites and the tensile test. Reproduced with permission from Reference [73]. Copyright 2011, Elsevier [Color figure can be viewed at wileyonlinelibrary.com]

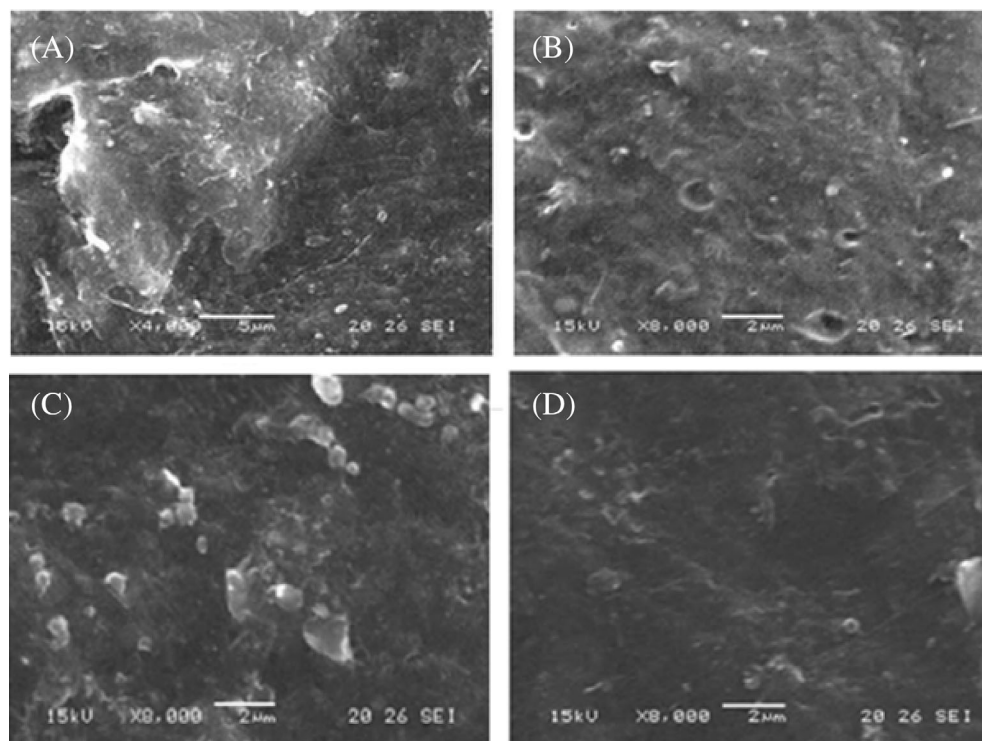


FIGURE 5 Scanning electron microscopy (SEM) micrographs of polypropylene (PP) nanocomposites with (A) and (B) 1 wt%, (C) 1.5 wt% and (D) 2 wt% graphene nanosheets (GNs) content at high magnification. Reproduced with permission from Reference [48]. Copyright 2012, John Wiley and Sons

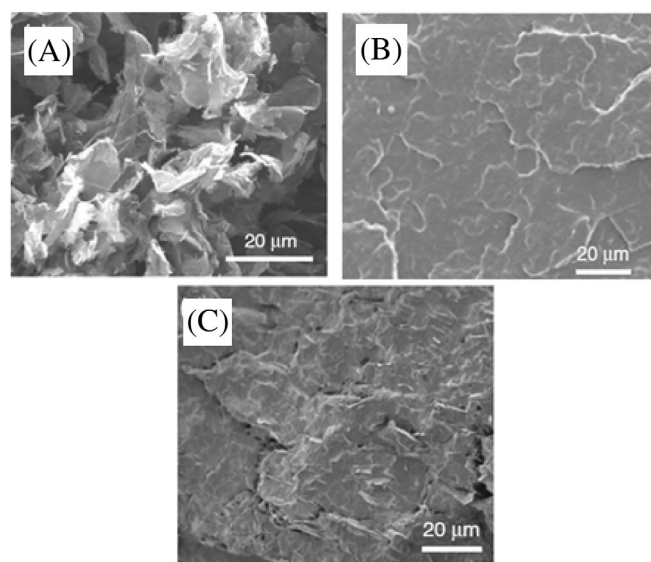


FIGURE 6 Scanning electron microscopy (SEM) micrographs of (A) the graphene nanoplatelets (GNP), (B) a fracture surface of the polypropylene (PP) and (C) a fracture surface of the PP/GNP nanocomposite with 5.2 vol% GNP loading. Reproduced with permission from Reference [76]. Copyright 2016, Elsevier

groups were eliminated, allowing PP latex to be readily coated with graphene because the residue of carboxyl of RGO interacted with the carboxyl of PP latex via H-bonding. Figure 9(C),(D) proved that almost all graphene sheets were fully exfoliated into mono sheets with small quantity of few-layers was noticed as pointed by the red

arrows.^[73] In addition, Quiles-Díaz et al.^[79] also found the similar observation where the graphene layers were partly exfoliated and bent in the iPP substrate due to the elongational and shear forces during the melt mixing process.

The combination of SEM and TEM results are important because both results able to support each other's to show stronger evidences as seen in Figure 10. Figure 10 (A) shows that most of the colloidal particles (PP latex) were in spherical shape and bowl-like particles resulted from water evaporation process. The spherical shape was remained after the PP latex was heated up to 95°C for 4 h as shown in Figure 10(B), but the spherical shape was replaced by wrinkled RGO networks as seen in Figure 10 (C),(D) after reduction by hydrazine. On the other hand, Figure 10(E),(F) shows comparison of morphology for samples added with and without PP latex. No aggregations was seen in RGO/PP latex composites sample as shown in Figure 10(E), suggesting that RGO nanosheets were homogeneously dispersed in PP matrix. In addition, no RGO was stretched out, signifying that RGO nanosheets were interacted strongly with the surface of colloidal PP. This strong adhesion was the result of interactions between the oxygen containing functional groups on RGO surfaces and the carboxyl molecules on MA-g-iPP chains.^[74] In other words, the RGO interacted strongly with the PP matrix more than the interaction with each other to form big particles. However, GNs showed graphite-like structure in Figure 10(F), indicating

FIGURE 7 ESEM micrographs of fracture surface of (A) 1 vol% xGnP-15/PP (scale bar 500 μm) and (B) 1 vol% xGnP-1/PP (scale bar 40 μm). Reproduced with permission from Reference [78]. Copyright 2007, Elsevier. GnP/PP, graphene nanoplatelet/polypropylene

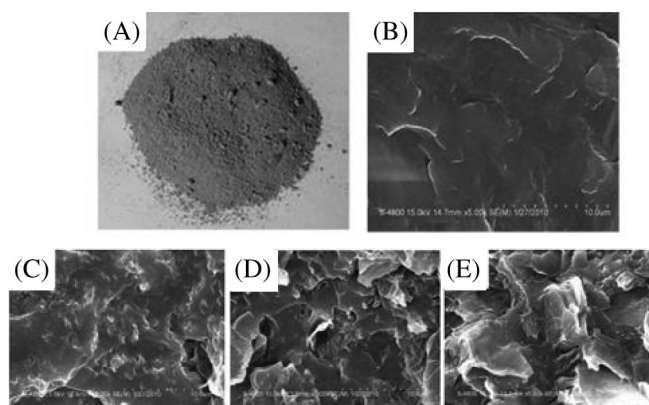
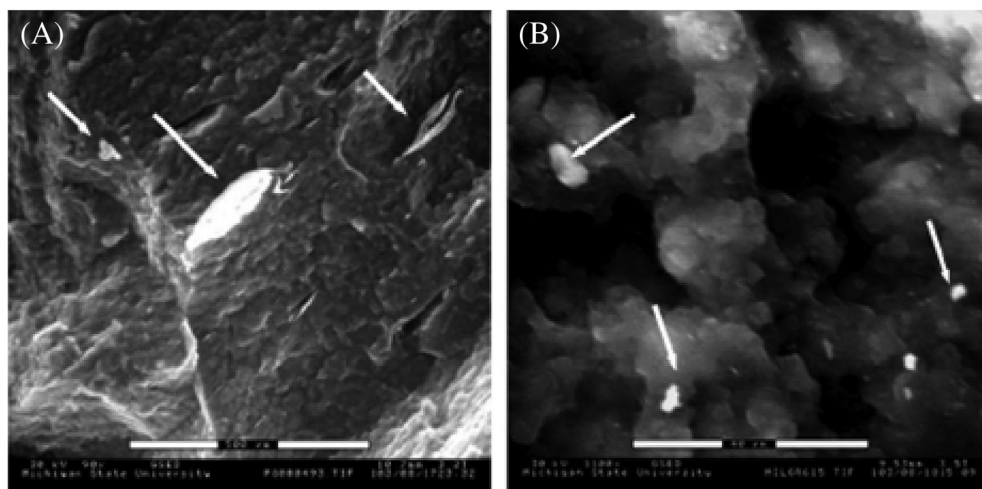


FIGURE 8 (A) Polypropylene/graphene oxide (PP/GO) nanocomposites powder as obtained after coagulation in ethanol (containing 0.1 M HCl). (B) Scanning electron microscopy (SEM) micrographs of fracture surface of neat PP. (C)–(E) SEM pictures of fracture surface of composites with GO loadings of 0.42, 1.52, and 4.90 wt%, respectively. Reproduced with permission from Reference [67]. Copyright 2010, American Chemical Society

that those nanosheets restaked with each other after reduction process. This occurred due to π - π stacking after oxygen-containing functional groups were eliminated.^[20,74]

3.2.2 | X-ray diffraction

X-ray diffraction (XRD) is one of the important characterization tools to investigate crystallinity, crystallite size, orientation of the crystallites, phase composition, and others.^[80] The investigation of nanofillers is as important as composite by using XRD. Therefore, characterization of graphite, GOs, and functionalized graphene sheets (FGs) were demonstrated by Qiu et al.^[81] A peak at $2\theta = 26.6^\circ$ is corresponded to graphite peak, with a d-

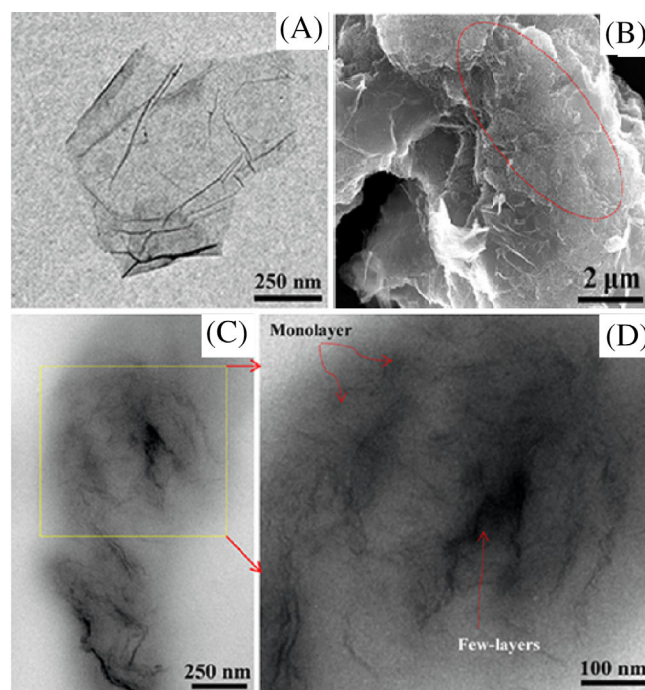


FIGURE 9 (A) Transmission electron microscopy (TEM) of graphene sheets, (B) scanning electron microscopy (SEM) micrographs of polypropylene (PP)-coated graphene and the PP latex was circled by red circles, (C),(D) TEM micrographs of graphene dispersion in PP matrix. Reproduced with permission from Reference [73]. Copyright 2011, Elsevier [Color figure can be viewed at wileyonlinelibrary.com]

spacing of 0.335 nm, as shown in Figure 10. Upon oxidation, the peak was shifted to $2\theta = 10.8^\circ$ with an increased d-spacing of 0.818 nm.^[81] Furthermore, no significant sharp peak was seen for FG samples in Figure 11. This reveals that FG was exfoliated well,^[81] with the possible formation of single sheets.^[30] The changes of characteristic peaks of PP and composites are important for

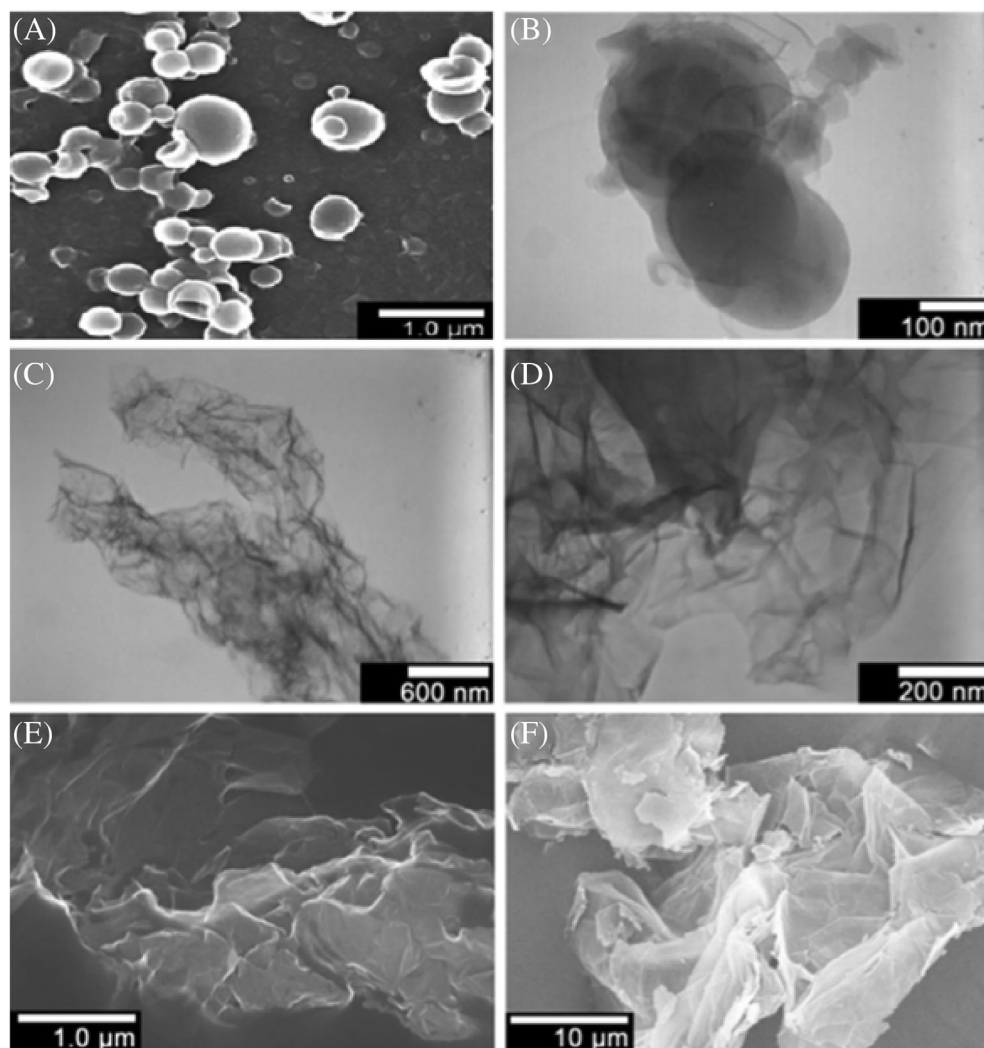


FIGURE 10 (A) Scanning electron microscopy (SEM) and (B) transmission electron microscopy (TEM) micrographs of polypropylene (PP) latex. (C),(D) TEM micrographs of the reduced graphene oxide (rGO)/PP latex composite ($f = 0.2$ vol%) dispersed in water before filtration. SEM micrographs of (E) fracture surface of the rGO/PP composite ($f = 0.2$ vol%) after hot-press molding and (F) agglomerated rGO nanosheets. Reproduced with permission from Reference [74]. Copyright 2013, Elsevier

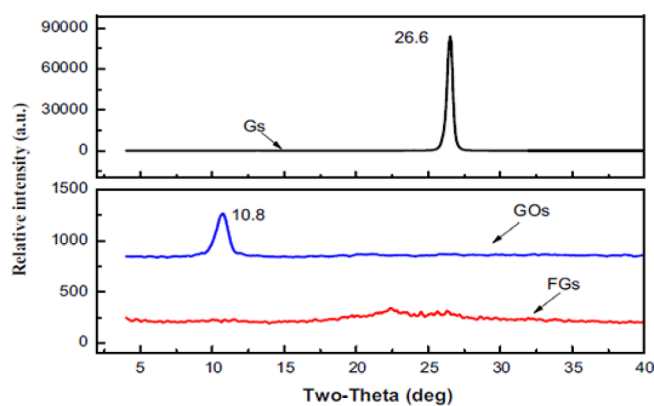


FIGURE 11 X-ray diffraction (XRD) spectra for graphite flakes (Gs), graphene oxide (GOs), and functionalized graphene (FGs). Reproduced with permission from Reference [81]. Copyright 2015, Elsevier [Color figure can be viewed at wileyonlinelibrary.com]

examination as shown in Figure 12, reported by Achaby et al.^[48] Neat PP displays several characteristic peaks at $2\theta = 14.1^\circ, 16.9^\circ, 21.2^\circ, 21.9^\circ, 25.5^\circ$, and 28.6° ,

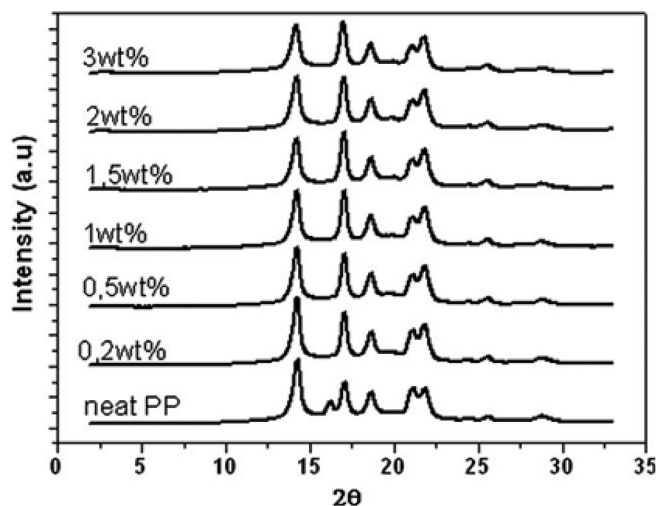


FIGURE 12 X-ray diffraction (XRD) spectra for graphene nanosheets-polypropylene (GNs-PP) composites, with reference to neat PP. Reproduced with permission from Reference [48]. Copyright 2012, John Wiley and Sons

respectively, representing (110), (040), (130), (111), (131) + (041), (150) + (060), and (220) of α -crystallization form of PP.^[82] β -crystallization form can be observed as well, with the presence of low intensity peak at $2\theta = 16.2^\circ$.^[83] Upon addition of GNs with PP, two major changes can be observed. First, no graphitic structure characteristic peaks were observed, indicating that GNs were dispersed successfully into the matrix.^[48] Second, the peak of β -crystal form that supposed to appear at $2\theta = 16.2^\circ$ was no longer presented, proving that β -crystal form was transformed to α -crystal form. This implies that the presence of GNs affects the crystal formation in PP matrix.^[48]

On the other hand, Ahmad et al described that the characteristic peak of GNP was seen as well in addition to the peak of PP (040).^[76] The peak of GNP (002) can be seen at $2\theta = 26.5^\circ$ and with every increment of GNP loadings, the intensity of peak increased as well. The presence of this peak proved that GNP particles remain their crystalline form in nanocomposites after the melt mixing was completed.^[84] Moreover, the formation of β -crystal form of PP (300)^[77] due to the nucleation caused by the addition of GNP was seen at the low angle side of the α -crystal form of PP peak (040) at higher loadings of GNP in composite.^[76] The same observation was reported by Jun et al^[85] and Zhang et al.^[86] In Figure 13(A), Zhang et al^[86] demonstrated that Gra/CaPi which acted as nucleating agent induced the formation of β -phase in iPP. Therefore, the spherulite size of crystal of iPP was reduced whereas high quantity of needle-like structure was observed in Figure 13(C). On the other hand, to avoid the reduced graphene from stacked or aggregated together again, PP latex can be added with GO first to form PP-coated graphene. However, no diffraction peak can be seen for PP latex and PP-coated graphene, concluding that PP latex was in amorphous state. In addition, graphene was well exfoliated into single sheets and

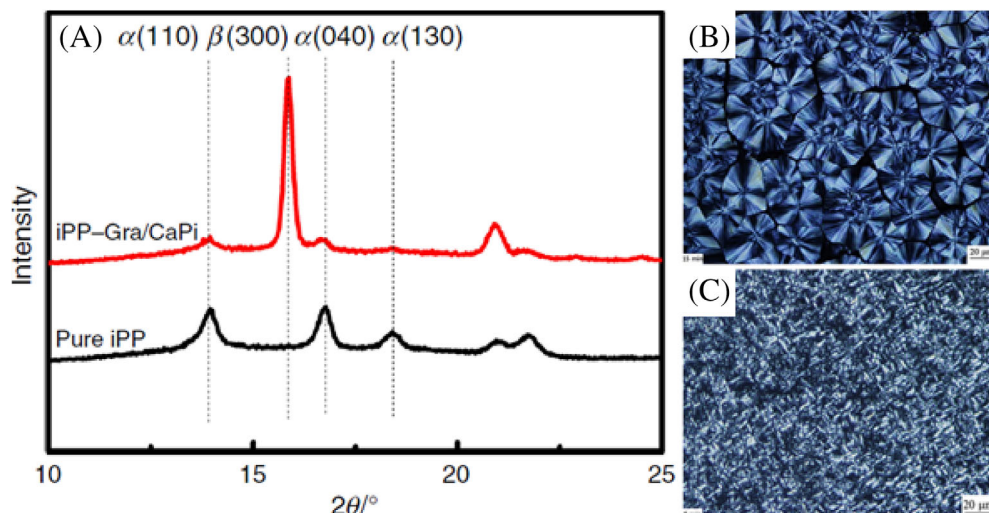
well-distributed in PP matrix since no characteristic peaks for graphene can be observed. The observed diffraction peaks were due to crystalline structure of PP matrix.^[73] Furthermore, Bafana et al^[87] also reported the addition of nanofiller did not influences the crystalline structure of PP. Similar pattern was reported by Wang et al. where the produced RGO/PP composites showed similar XRD pattern with no characteristic broad peak of RGO at $2\theta = 24^\circ$ but the presence of PP peaks was noticed.^[74] Besides, the addition of graphene sheets which caused the changes in crystal faces of (110), (040), (130) was reported as well, indicating the presence of graphene sheets alters the crystallization behaviors of the matrix.^[73]

The addition of nanofiller in the polymer matrix can induces the nucleation to some extent depends on the properties of nanofiller. Since the peak intensity ratio of $\alpha(040)/\alpha(110)$ represents the nucleating efficiency of α -form, Patchanee et al^[50] demonstrated five different nanofillers namely GNP, rGO, pGO, PP-modified GO (PP-GO) where the GNP showed the highest nucleation efficiency, followed by $rGO > pGO > PP-GO$. The wrinkles and buckling structures of rGO are the reason of lower nucleation efficiency than GNP. Furthermore, the presence of residue of oxygen functional groups reduced the nucleation efficiency of pGO and PP-GO. For the comparison between pGO and PP-GO, higher orientation degree was noticed for pGO than PP-GO in spite of same reduction process. Thus, the coverage of PP-OH at GO surfaces decreased the nucleation efficiency in PP matrix.

3.2.3 | Thermogravimetric analysis

Thermogravimetric analysis (TGA) is another characterization tool to examine the thermal stability of material.

FIGURE 13 Comparison between pure isotactic polypropylene (iPP) and iPP nucleated with 0.3% mass Gra/CaPi for the (A) WAXD curves (B) Pure iPP: 15 min (C) nucleated iPP: 3 min. Reproduced with permission from Reference [86]. Copyright 2019, Springer Nature [Color figure can be viewed at wileyonlinelibrary.com]



TGA is used to verify both nanofiller loading and to investigate the effect of nanofiller loading on the thermal steadiness. This test also able to further analyze the presence of functional groups in carbonaceous materials.^[88] With the addition of GNs into the PP matrix, the increments of thermal stability of $T_{5\%}$, $T_{15\%}$, and $T_{\max}$ were noticed by roughly 40, 20, and 10°C, respectively. Therefore, the TGA results proved that the thermal stability for the composites was improved with increment in GNs loadings.^[48] It is because the resistance of thermal deterioration was enhanced by preventing the movement of unstable molecules inside the nanocomposites which very much affected by GO-PP chains interactions. In addition, the derivative thermogravimetry (DTG) peak became broader at GNs loadings higher than 1.5 wt% due to mechanism and reaction kinetics effects. This allows PP to be widely used for applications require high thermal stability.^[48] Besides, GNP also increased the degradation temperature of PP/GNP by 70°C as compared with neat PP when 1 wt% of GNP was added.^[76] Furthermore, as the filler loading increased to 10 wt%, the $T_{5\%}$ was increased to ~427°C from ~298°C (neat PP). Therefore, a significant improvement of degradation temperature was affected by an increase of filler loadings.^[76] Besides, higher surface area of GNP also influenced the efficiency of enhancement of degradation temperature of composites.^[89] Several comparable findings about the effect of GNP in enhancements of thermal stability were found in LLDPE,^[90] HDPE,^[91] polylactide^[84] and PMMA.^[92]

3.2.4 | Raman spectroscopy

The Raman spectroscopy is a characterization tool to investigate the sample imperfection, such as CNTs and graphene.^[93] In the spectra, two common peaks that can be found are D and G peaks. D peak refers to a disorder-induced Raman mode and is associated with the transverse optical (TO) branch near the K-point in the Brillouin zone.^[94] The peak is formed due to the breathing modes in C—C ring structure and the amount of disorder structure in graphene can be determined from its intensity.^[95] On the other hand, G peak is formed due to the doubly degenerated E_{2g} phonon mode at the Γ point at the Brillouin zone center,^[96] which is corresponded to C (sp^2)—C (sp^2) bond stretching vibrations near to 1580 cm^{-1} .^[94] The intensities of both peaks can be used to estimate the quality of the produced graphene and graphene-based materials by calculating the ratio of I_D/I_G .^[97] Therefore, it is interesting to study the influences of nanofillers to the composites by using Raman spectroscopy. The neat PP showed a small peak near 3000 cm^{-1} and the intensities of D, G and 2D bands were not

influenced with the increasing in volume percentage of the GNP filler.^[76] It was due to the smaller spot size of Raman laser than the sizes of particles. Besides, no favored direction of the GNPs was obtained at different spots using polarized Raman spectroscopy.^[98,99] In addition, the presence of asymmetric shape of 2D band indicates that GNPs consist of multilayer graphene.^[76]

Furthermore, Raman spectra of nanofillers and composite for the purpose of comparison were reported by Song et al.^[73] (Figure 14). For graphite oxide, both D and G peaks appeared to be wider and slightly red shifted with an increase in I_D/I_G value than graphite. Interestingly, after reduction, both D and G peaks for graphene shifted to the similar peak locations of graphite. Moreover, the D and G bands of graphene showed much wider than graphite and I_D/I_G value was higher than that of graphite oxide which explained by an increase in amount of defect structure of graphene after reduction.^[100] In addition, the I_D/I_G value of PP-coated graphene was lower than graphene due to the presence of PP latex which influenced merely the effectiveness of reduction.^[100]

Besides, the effectiveness of interaction between filler and polymer can be examined from the characteristic of G band. It was showed that the tensile strain reduces G band frequency due to the extension of carbon-carbon bonds whereas compressive strain increases the G band frequency.^[101] Thus, the G band frequency is changed

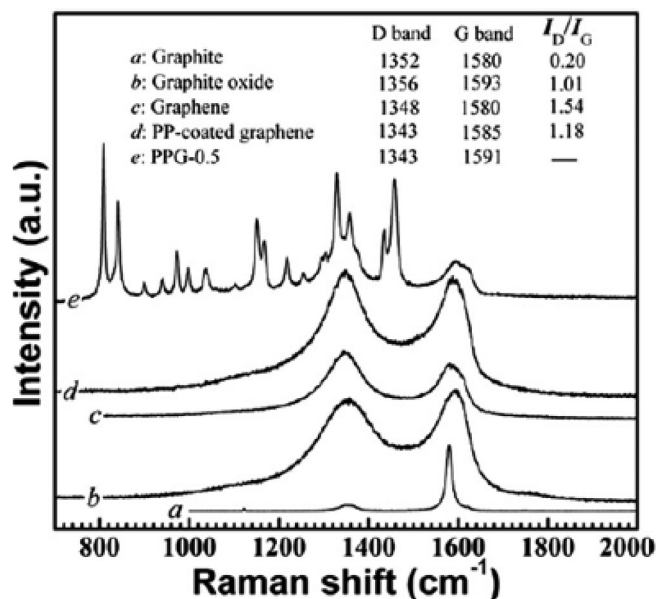


FIGURE 14 Raman spectra for (A) pristine graphite, (B) graphite oxide, (C) graphene, (D) polypropylene (PP)-coated graphene and (E) PP/graphene nanocomposites with 0.5 wt% graphene loading. Reproduced with permission from Reference [73]. Copyright 2011, Elsevier

according to the stretching or compressing of composite in mechanical test by the effect of interfacial stress transfer.^[102,103] A blue shift in G peak was reported which attributed to the shrink stress during the crystallization of MA-g-iPP chains. Furthermore, the shift also proved that strong interface interactions existed between RGO and PP matrix.^[74] Raman ratio (I_D/I_G) for RGO and RGO/PP were determined to be 1.27 and 1.55, respectively. The increment in I_D/I_G confirmed that the amount of defect structures was increased.^[73] Therefore, the increment ratio also indicates that the efficiency of GO reduction was slightly reduced with the presence of PP latex.^[74]

3.2.5 | X-ray photoelectron spectroscopy

X-ray photoelectron spectroscopy (XPS) was used to investigate the changes in chemical bonds before and after functionalization of ramie fiber (RF) by using GO.^[104] As seen in Figure 15, the peak intensities of C—OH and COOH were reduced whereas the peak intensity of C—O—C/C=O was increased. The reaction between oxygen-containing functional groups from GO and RF to produce new bonds, namely ether and ester bonds are related to the reason of opposite patterns of peak intensities. Therefore, the presence of new bonds improve the interfacial interaction between GO and RF. The wrinkle shape of GO create interlock effect where the interfacial layers movement between GO and RFs are inhibited.^[105,106]

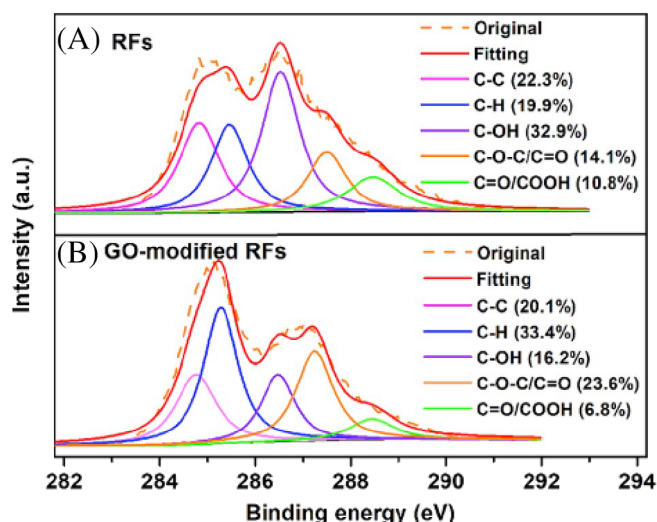


FIGURE 15 X-ray photoelectron (XPS) spectra of C1s of (A) RFs and (B) graphene oxide (GO)-modified RFs. Reproduced with permission from Reference [104]. Copyright 2019, Elsevier [Color figure can be viewed at wileyonlinelibrary.com]

3.2.6 | Mechanical properties

Mechanical properties of composites can be enhanced with addition of nanofillers. The improvement in tensile strength is an indicator of well dispersion of nanofillers into the polymer matrix. This improvement can be achieved by the presence of strong interfacial interaction between polymer matrix and fillers. However, in this section, some research works where the nanofillers were mixed with PP without any functionalization were demonstrated. For example, the additional of GNs as nanofiller into PP matrix showed improvement in the mechanical properties of the produced composites. The same observation was also reported by Mistretta et al.^[107] Interestingly, with only 1 wt% of GNs was added, an increase of about 64% and 54% for both Young's modulus and the tensile strength, respectively, were shown significantly when compared with neat PP values. The Young's modulus and tensile strength achieved almost 100% and 81%, respectively, when the loading of GNs was increased to 3 wt%. On the contrary, elongation at break of the composites showed great decrement for GNs loading as low as 0.2–0.5 wt%. The high aspect ratio, strong interlayer interaction, and high intrinsic mechanical properties of GNs are the factors for the improvement in interlayer stress transfer by limiting the movement of polymer chains.^[48]

Besides GNs, GNPs also show improvement in Young's modulus from 1.3 to over 2.0 GPa when increased GNP loadings from 0.5 to 5.2 vol%^[76] and the results revealed that the nanocomposites were more stiff as more filler loadings were added. However, the fracture strain decreased from 250% for neat PP or 0% GNP loading to less than 10% for 5.2 vol% GNP loading. This pattern indicated that the produced nanocomposites were more brittle than neat PP. For yield stress, slight reduction from 34 to 30 MPa was observed. There are three factors were elucidated: first, the GNPs are thick and thus, reduces the Young's modulus of nanocomposites as compared with monolayer graphene. Second, the random orientation of GNPs decrease the efficiency of reinforcement by a factor of 8/15 and third, the short nanoplatelets fail to reinforce composites as compared with longer ones, implying if this relevant to the 15 μm GNPs in their study, it causes 80% of reduction in the effective Young's modulus in the region of 100 GPa.^[76]

To avoid the RGO from stack or aggregate to form big sizes particles, Song et al.^[73] applied PP latex to coat it onto GO surface before the reduction of GO using hydrazine hydrate. However, the presence of PP latex caused a decrease in all mechanical parameters as compared to neat PP. With only 4 wt% PP latex, $\sim 20\%$, $\sim 50\%$ and $\sim 60\%$ of reduction in tensile strength, Young's

modulus and elongation at break were observed respectively. The reduction in tensile strength and Young's modulus were caused by low mechanical strength of PP latex and plasticization effects. For elongation at break, the additions of polar surfactant and additives were the main factors of the decreased in the elongation. Therefore, an increase in graphene loading also increased the amount of the polar surfactant and additives, thus caused more effects on the mechanical properties.^[73]

Comparison between different types of nanofillers to the mechanical properties of composites is very interesting because nanofillers have different physical and chemical properties. Table 2 displays the tensile properties for the produced PP composites (PP/GNP, PP/rGO, PP/pGO, and PP/PP-GO) where enhancement in both tensile strength and Young's modulus was noticed when 1.0 wt% of graphitic filler was added into PP matrix. However, when the filler content was further increased up to 3.0 wt %, tensile strength showed small decrements, possibly caused by poor dispersion. An increase of layer density due to exfoliation of rGO resulted higher tensile strength for PP/rGO as compared with PP/GNP. Furthermore, PP/pGO had the lowest tensile strength, which was due to the presence of oxygen-containing functional groups that lead to poor compatibility and poor interaction with PP matrix.^[50] Among all composites, PP/PP-GO (1.0 wt %) showed the highest tensile strength with 28% improvement from pristine PP. This improvement was defined as high because no compatibilizer was used and no chemical modification was done on the graphene layers. Those composites with compatibilizer resulted 35% and 29% enhancement when PPMA was grafted onto diamine and amine-modified GO, respectively.^[94,117] High tensile strength value as shown by PP/PP-GO proved that better dispersion and interfacial interaction were achieved by PP-GO with PP matrix as compared to rGO and pGO and

confirming that PP-OH were successfully grafted onto GO. However, PP/PP-GO showed decrement in elongation at break when PP-GO loading was increased from 1.0 to 3.0 wt%.^[50] This trend was due to either non-homogeneous dispersion or loss of plasticity which resulted from limitation of platelet that reduces mobility of lamellae from being stretched and orientated during tensile test.^[108]

3.3 | Synthesis of AGO and AGO/PP composite

The direct mixing of GO and PP by using twin-screw extruder or other mixer is not the proper method due to the agglomeration issue of GO after mixed two incompatible materials together. The structure of GO is different with pristine graphene due to the presence of oxygen containing functionalized groups at its basal plane and the edges as shown in Figure 16.^[109] The presence of these variety of oxygen functional groups allows GO to be easily dispersed in aqueous media due to hydrophilic characteristic. On the contrary, PP behaves hydrophobic and incompatible with GO.^[109] Therefore, the improper mixture of GO and PP could cause the failure to improve the mechanical properties of the produced composites.

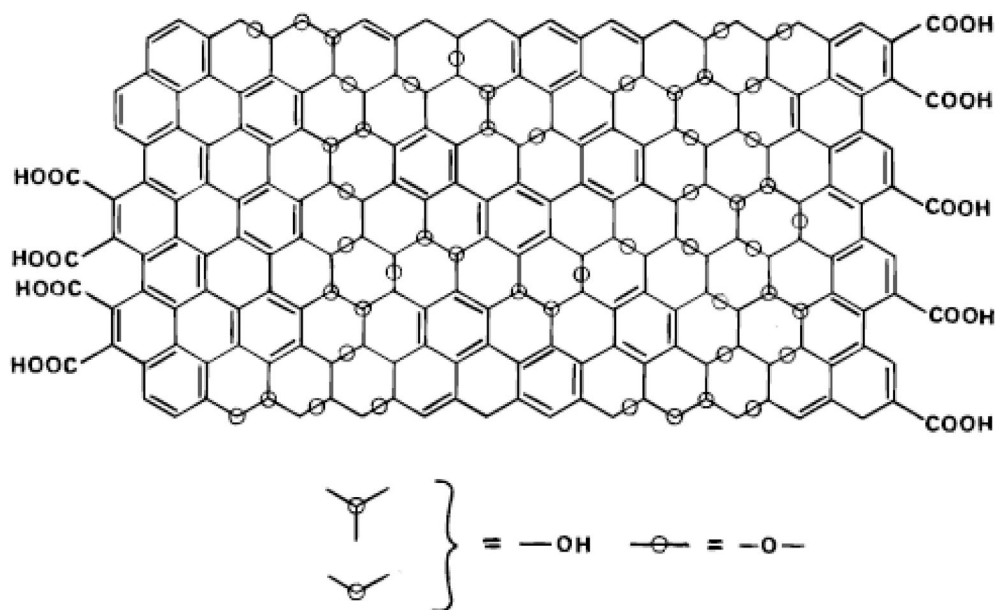
One way to solve this poor mixing due to agglomeration issue, the GO must be alkylated prior to the mixing with PP. Alkylation is a process where additional alkyl groups are introduced to the target such as GO to create larger interaction sites with the substrates such as PP to achieve homogenous mixing for improvement of mechanical properties.^[110] One example of alkylation was reported by Klunbud et al where GO nanosheets were decorated with amino-silane modified silica nanoparticles as reinforce filler for composites.^[111]

TABLE 2 Tensile properties of PP composites

Sample	Tensile strength (MPa)	Young's modulus (MPa)	Elongation at break (%)
Pristine PP	30.2 ± 0.4	470 ± 15	>300
PP/GNP (1.0 wt%)	36.3 ± 1.1	550 ± 22	>300
PP/GNP (3.0 wt%)	35.1 ± 0.7	567 ± 14	38 ± 8
PP/rGO (1.0 wt%)	36.4 ± 0.4	554 ± 24	>300
PP/rGO (3.0 wt%)	36.2 ± 0.6	552 ± 40	62 ± 14
PP/pGO (1.0 wt%)	35.3 ± 0.6	516 ± 29	>300
PP/pGO (3.0 wt%)	35.3 ± 0.5	549 ± 25	79 ± 12
PP/PP-GO (1.0 wt%)	38.7 ± 1.0	562 ± 34	>300
PP/PP-GO (3.0 wt%)	37.1 ± 0.5	564 ± 47	>300

Note: Reproduced with permission from Reference [50]. Copyright 2016, Multidisciplinary Digital Publishing Institute. Abbreviations: GO, graphene oxide; GNP, graphene nanoplatelet; PP, polypropylene; rGO, reduced graphene oxide.

FIGURE 16 Chemical structures of graphene oxide (GO) with epoxy and hydroxyl groups at basal plane and carboxylic groups at the edges. Reproduced with permission from Reference [109]. Copyright 2014, Elsevier



The interfacial interaction of GO and amino-silane functional groups was done through an electrostatic interaction between the negative charge and the positive charge of oxygen-containing groups of GO and amino-silane functional groups, respectively at the surface of silica nanoparticles. With the presence of GO, the dispersion of SiO_2 nanoparticles onto the polymer matrix was easier because of its hydrophilicity property with the attachment of oxygen functional groups.^[111] The GO nanosheets decorated with amino-silane modified silica nanoparticles could be used with another polyolefins polymers, not limited to PP only.

The modification of GO surface does not limit to amino-silane only where Hao et al produced alkylated GO (namely AGO), by conducting reaction between GO and octadecyl isocyanate ($\text{CH}_3[\text{CH}_2]_{17}\text{NCO}$) using DMF as medium. The alkylated chains ($-\text{CONHC}_{18}\text{H}_{37}$) were covalently grafted in a reaction between carboxylic acid groups ($-\text{COOH}$) of GO with $\text{CH}_3(\text{CH}_2)_{17}\text{NCO}$.^[112] Recently, polyoxyethyleneoctylphenyl-10 (OP-10) was used in preparing PP/GO composite membranes.^[113] Huang et al^[114] and Bagheri et al^[115] proposed that by introducing pores (or void) with micrometer sizes to alter the mechanical properties of composites. In this production, OP-10 acts as a pore-forming agent with the presence of methyl at one side and hydroxyl at the other side. The addition of OP-10 enhanced the dispersion of GO and induced the formation of many pores during tensile test process in the stretched PP/OP-10 sample simultaneously.^[113]

An interesting research was conducted by Ryu and coworkers about the effects of different lengths of long chain alkylamines, such as hexylamine (A-6), decylamine

(A-10), hexadecylamine (A-16), and octadecylamine (A-18) on the melt crystallization and properties of PP nanocomposites.^[94,116] There are two types of reactions that are expected to occur. First reaction is amidation reaction between amine groups and carboxylic acid which appeared at sites of GO. Another reaction is nucleophilic substitution that occurs between amine and epoxy groups at the surface of GO. These reactions could happen easily with the presence of oxygen-containing functional groups at the GO surface,^[33,94] which is beneficial as compatibiliser for mixing between alkylamines grafted GO with polymer matrix.^[94] In addition to long chain alkylamine, hexamethylene diamine (HMDA) was also used to synthesize diamine-functionalized GO. In this research, HMDA was priority important to improve the dispersion and interfacial strength before the produced HMDA-modified GO were melt blended with PP. The chemical grafting reactions of HMDA onto GO sheets were achieved via two types of reactions. One of the reactions is amidation reaction between amine groups and carboxylic acid sites of GO, same reaction that occurs to the reaction between alkylamine and GO. Another reaction is the nucleophilic substitution reaction between amine and epoxy groups on GO surface.^[117]

The addition of compatibilizer could facilitate the mixing between nanofillers and polymer matrix before the melt blending to attain the well dispersion of nanofillers in polymer matrix. For example, maleic anhydride-grafted-polypropylene (MA-g-PP) was used as compatibilizer in the melt mixing of various amounts of AGO loadings with polypropylene matrix to form composites.^[117] The possible mechanisms for reaction between MA-g-PP and HMDA are nucleophilic

substitution reaction between epoxy and alkylamines occurs in the basal plane meanwhile amidation reaction happens at the edges of GO layers.^[117] Besides, another research about modification of GO with *p*-phenylenediamine (PPD) and cyanuric chloride (CC) to prepare functionalized GO (fGO)/MA-g-PP composites was also reported where the MA-g-PP was used as compatibilizer.^[108] By using this method, fGO was well dispersed and showed good interaction with MA-g-PP. The purpose of CC was to add more reactive points for interaction between GO and MA-g-PP.^[108] Apart from the use of alkylamines, Yun et al. presented a work regarding application of tetra-*n*-octylammonium bromide (TOAB) and dodecyl iodide to produce alkylated GO (AGO).^[118] In this work, PP and produced AGO were dispersed in xylene separately before mixing together to form AGO/PP nanocomposites. Besides, Hidayah et al. also^[119] applied TOAB to functionalize GO to prepare AGO and then, it was further added with PP using twin screw extruder.

Rather than to conduct the modification on the surface of GO, Sánchez-Valdes et al.^[120] demonstrated the functionalization of PP prior to the mixing with GO. They reported the use of MA-g-PP to react with 2-(2-(dimethylamine)-ethoxy) ethanol (DMAE) in the melt condition to form amine-alcohol functionalized polypropylene (PPgDMAE) which believed to be more effective in improving the dispersion and interfacial interaction between fillers and polymer matrix. Interestingly, this work compared the performance of PPgDMAE with GO by using two different processing methods, namely melt and solvent mixing. The grafted PP with amine alcohol compatibilizer contained amine polar groups which interacted strongly with oxygenous polar groups of GO. Besides, epoxide groups of GO also could react with amine polar groups in the ring-opening reaction.^[121] Furthermore, the interaction between carboxylic acid of GO and amine polar groups formed the amido linkage during melt mixing.^[122] Therefore, they found that solvent mixing enhanced the mechanical thermal stability and electrical properties as compared with melt mixing. Besides HMDA, PPD, and DAME, hydrophilic bis(3-aminopropyl)-terminated poly(ethylene glycol) (NH₂-PEG-NH₂) was used to covalently functionalize GO (GO-PEG).^[123] GO-PEG was synthesized by attaching NH₂-PEG-NH₂ onto the GO surface. This would allowed the MA-g-PP (used as a compatibilizer) to react with GO-PEG which formed amide groups, creating PP-g-GO. Since PEG contains flexible backbone that appears in many complex shapes,^[124] it induces the interfacial interactions with PP oligomer chains.^[125] Then, PP-g-GO/MA-g-PP was mixed with PP to produce PP-g-GO/MA-g-PP composite by melt mixing.

Besides MA-g-PP, octyltriethoxysilane (OTES) was also used as compatibilizer in preparing GO/PP nanocomposites to investigate the resistance of the produced nanocomposites toward scratch. OTES was added into the mixture of PP and GO before the mixture was subjected to 12 hours of stirring. The solvent was allowed to dry in a vacuum oven. Furthermore, the GO/PP treated with OTES was mixed using an internal batch mixer.^[126] Other than all methods as mentioned previously, solution mixing was also used. To do the solution mixing, xylene is one of the solvent that was used in some published articles such as Yun et al.^[104] In their work, AGO was dispersed homogeneously in xylene whereas PP was dissolved in xylene at 130°C under a nitrogen atmosphere in a reaction flask. The mixture of AGO and xylene was added into the mixture of PP and xylene to induce the reaction between AGO and PP under stirring for 30 min.

3.4 | Characterization of AGO and AGO/PP composites

3.4.1 | SEM/FESEM/TEM

SEM was used to study the interaction between GO-PP and alkylamine-modified GO-PP. At same loading of 1 wt %, serious aggregations can be seen at the surface of GO-PP samples, but the strong interfaces of alkylamine-modified GO-PP composites were spotted as well, indicating that long chain alkylamines was grafted successfully on GO.^[94] Besides, SEM images are very useful for confirmation of presence of any filler in matrix because smooth surface of pristine PP can be observed before addition of any filler. However, after the addition of GO, the surface of GO/PP was turned rougher clearly with the formation of GO in plaque form.^[126]

The comparison between melt and solvent mixing is important and interesting to help researchers to choose the suitable method to prepare composites. After melt mixing, aggregation of GO particles with size bigger than 10 µm can be observed at the composite surface whereas smaller aggregation size GO particles of 5 µm were formed in the composite which prepared using solvent mixing. The SEM results were further supported by the TEM micrographs where higher amount of exfoliated GO was dispersed successfully in the PP matrix, indicating that solvent mixing provided better dispersion of filler in polymer matrix than melt mixing.^[120] Besides, the mixing between GO and PP without functionalization resulted in a very poor dispersion of GO in PP matrix whereas poor dispersion and sizes of GO agglomerates were significantly reduced with the addition of OP-10.^[113]

It is because OP-10 is an amphiphilic pore-forming agent with methyl at one side and hydroxyl at the other side. Thus, with the attachment of functional groups on OP-10 and GO, the strong interaction between them leading to good adhesion with PP matrix.^[19]

SEM provides another important information about the samples morphologies after the mechanical test. A report about the anti-scratch properties of nanocomposites using SEM analysis was reported. Parabolic-shaped ripples can be observed on the surface of pristine PP and GO/PP nanocomposite. On the other hand, GO/PP/OTES nanocomposite had the formation of regular zigzag scratch tracks on its surface. This pattern implied that the GO/PP/OTES nanocomposites demonstrated more scratch resistance as compared to those nanocomposites without OTES. Voids and debris were noticed at the surfaces of scratched GO/PP nanocomposites where the presence of void was affected by disconnection between the GO and PP. However, this phenomenon was not seen in the scratched GO/PP/OTES nanocomposite due to its higher elastic recovery than that of GO/PP nanocomposites without OTES. Moreover, the increasing loading of GO and OTES in the produced nanocomposites enhanced the scratch resistance in which the elastic recovery was improved by 75.1% as compared with pristine PP with 51.1% elastic recovery. Therefore, the increment in GO loadings decreased the damages possibility at the surface of OTES treated samples than that of pristine PP and GO/PP without OTES nanocomposites.^[126]

Energy-dispersive spectroscopy X-ray was able to trace the element distribution on a sample surface and it was demonstrated by Anastacio-López.^[127] They investigated the successful dispersion of Ca^{2+} at the GO surface due to this metal was applied as a linker to connect pimelic acid and oxygen containing functional groups of GO (Figure 17). In addition, they also showed that the absence of Ca^{2+} resulted in the unstable attachment of pimelic acid to the GO surface and caused it to be washed away easily.

3.4.2 | X-ray diffraction

The characterization of XRD able to investigate the interlayer distance between graphite sheets and appearance of new compound after grafted with functional groups. After the functionalization of GO with PPD, two new peaks can be observed at 6.3° (d-spacing = 1.403 nm) and 24.2° (d-spacing = 0.367 nm). The d-spacing was increased from 0.859 to 1.403 nm implied that the PPD was intercalated and grafted onto GO.^[128] Besides, the appearance of peak at 24.2° proved the successful reduction of GO by PPD. The d-spacing (0.367 nm) for peak at

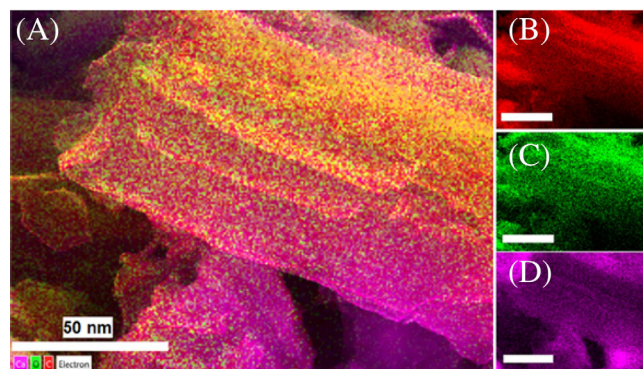


FIGURE 17 Scanning electron microscopy/energy dispersive spectroscopy (SEM/EDS) observations: (A) Image of graphene oxide (GO)-PA, (B) presence of carbon, (C) presence of oxygen, (D) distribution of calcium. Reproduced with permission from Reference [127]. Copyright 2012, The Royal Society of Chemistry [Color figure can be viewed at wileyonlinelibrary.com]

24.2° was merely higher than the d-spacing for pristine graphite (0.335 nm) due to the existence of residue functional groups on GO-PPD layers.^[129] For fGO, only broad peak at 25.2° can be seen. However, after MAPP was modified chemically onto the grafted functionalized graphene oxide (MAPP-g-fGO), several peaks of PP were detected only without any nanofiller peak was seen.^[108] A small peak at the shoulder of (040) peak was noticed which corresponded to (300) plane of β -crystal,^[130] implying that fGO behaves as a β -nucleating agent for PP crystallization at fGO loading of 1 wt% only. On the other hand, Bo et al.^[131] noticed that GO showed strong α -nucleation effect which created negative effect to the β -crystallization of β -iPP/GO composites. Therefore, they added the β -NA as β -nucleating agent to counteract with the GO to optimize the crystallization and morphology of composites.^[108]

The examination of the successful covalent functionalization of GO with MAPP chains is important by using XRD before the preparation of composite. After modification of GO with NH_2 –PEG– NH_2 , GO-PEG displayed a diffraction peak at 6.1° , downshifted from 9.1° .^[132] Its interlayer spacing also increased to 14.5 Å as compared with GO (9.7 Å) due to the attachment of NH_2 –PEG– NH_2 . The characteristic peaks of NH_2 –PEG– NH_2 at 19.5 and 23.6° were seen in GO-PEG spectrum which also reported by Petit et al.^[133] Furthermore, in PP-g-GO spectrum, a new broad peak at around 17.5 – 25° was observed, indicating MAPP was grafted onto GO-PEG surface in the reaction with amino groups. Besides, MAPP also acts as compatibilizer to induce solution blending with PP-g-GO.^[123]

An increase of interlayer spacing (d-spacing) can be noticed when longer chain lengths alkylamine were used.

Figure 18(A) shows the XRD spectra for GO and alkylamine-modified GO which were modified chemically with different chain lengths. In Figure 18(A), no peak was seen for hexylamine-grafted GO (GO-A6), confirming that the amount of graphene sheet piling was not affected by the addition of hexylamine.^[116] As the chain length of alkylamine increased, a broad peak can be observed in the 2–10° range except the GO-A6 where the shortest length of hexylamine has no effect to the amount of graphene layer stacking. Furthermore, the position of broad peaks were influenced by the presence of different chain lengths of alkylamine (Figure 18(A)). Decylamine-grafted GO (GO-A10), hexadecylamine-grafted GO (GO-A16), and octadecylamine-grafted GO (GO-A18) displayed a broad peak at $2\theta = 5.8^\circ$ ($d = 1.52$ nm), at $2\theta = 5.0^\circ$ ($d = 1.77$ nm) and at $2\theta = 3.7^\circ$ ($d = 2.39$ nm), respectively ((Figure 18(A) inset)).^[116] On the other

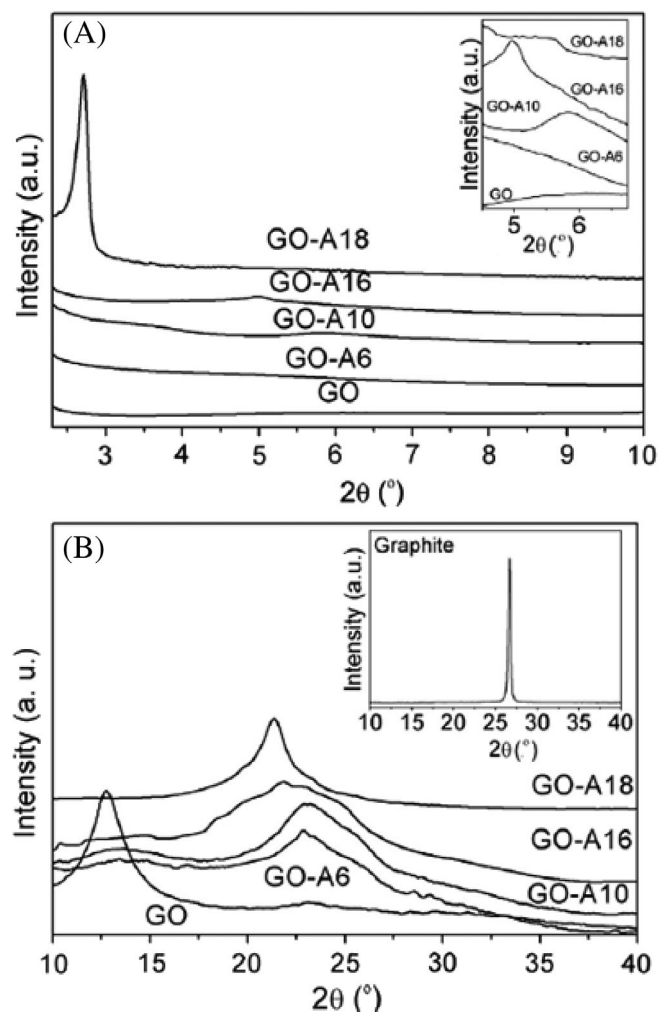


FIGURE 18 (A) X-ray diffraction (XRD) spectra of graphene oxide (GO) and alkylamine-grafted GO and (B) wide angle XRD spectra of GO and alkylamine-modified GO. Reproduced with permission from Reference [116]. Copyright 2013, Elsevier

hand, Figure 18(B) shows XRD spectra of pure GO and amine-modified graphene (GO-A6 to GO-A18) from 10° until 40°. A (002) peak was seen at $2\theta = 26^\circ$ which displayed for graphite sample with d-spacing of 0.34 nm ((Figure 18(B) inset)).^[116] After oxidation, the peak was shifted to $2\theta = 12.6^\circ$ with d-spacing of 0.7 nm, confirming that oxidation increased the d-spacing from 0.34 nm of pristine graphite.^[134,135] In addition, the modification of GO with alkylamines resulted in the disappearance of peak at $2\theta = 12.6^\circ$ of GO. However, broad peaks at around $2\theta = 21^\circ$ – 24° can be observed. The d-spacing was determined to be around 0.39–0.14 nm and the peak intensity was reduced as well.^[116] This pattern might be contributed by the reaggregation of graphene layers caused by the solvothermal reduction.^[135]

Besides, another types of alkyl chain called HMDA was used as alkylation agent and grafted onto GO which named as HMDA-modified GO (AGO). For GO, a peak can be observed at $2\theta = 10.0^\circ$ whereas AGO sample exhibited a peak at $2\theta = \sim 7.5^\circ$ equivalent to (002) diffraction plane. The peak position was shifted from 10.0° to 7.5° due to the intercalation of HMDA molecules between the GO sheets.^[117] Meanwhile, another broad peak can be observed as well at 2θ range 11– 25° with (001) diffraction plane. The presence of this peak validated the successful reduction of GO and elimination of intercalated water molecules and the oxide groups.^[136] Similar with GO, AGO displayed a peak at $2\theta = 42^\circ$ which assigned to (100) diffraction plane.^[117] On the other hand, the formation of α crystalline states in long-chain alkylamine-modified GO/PP composites was reported by Ryu and Shanmugharaj.^[94] They found an obvious increasing pattern in the intensity ratio between (040, $2\theta = 17.1^\circ$) and (110, $2\theta = 14.3^\circ$) was happened to alkylamine-modified GO-PP (PPGO-A10-1.0 and PPGO-A18-1.0) when compared to neat PP or GO-filled PP composites, indicating an increasing nucleating efficiency by alkylamine.^[137,138]

3.4.3 | Thermogravimetric analysis

The amount of functional groups in carbonaceous materials can be estimated by using TGA.^[88] It is because different material has its unique decomposition temperature range. GO started to show its weight loss at above 40°C . At temperature below than 170°C , almost 40% weight loss was noticed which caused by evaporation of adsorbed water. As the temperature increased from 170 to 260°C , another 20% rapid weight loss was observed which due to the elimination of oxygen containing functional groups from the GO.^[117] The weight loss in GO was happened significantly than that of HMDA modified

GO (AGO) from 40 to 260°C due to AGO had much lower amount of oxygen containing functional groups after the reduction of GO.^[117] In addition, from 220°C to 450°C, higher weight loss can be detected because of the deformation of HMDA molecules which grafted onto the GO.^[136] On the other hand, for alkylamine-modified GO, ~30% weight loss was noticed from 220 to 520°C, proving that the attachment of alkylamine increased the thermal stability of GO. Nevertheless, for temperature below than 100°C, all alkylamine-modified GO showed almost zero weight loss, implying an improvement of hydrophobicity which reduced the quantity of absorbed water than that of pure GO that usually showed weight loss ca. 10% below 100°C. At temperature below 400°C, the thermal resistance for alkylamine-grafted GO samples increased with increasing chain length of modified GO which also consistent with an increase of interlayer spacing between graphene sheets.^[116]

Yun's research groups investigated thermal characteristics for both fillers and composites. The weight loss for AGO was approximately 22% more as compared with GO^[118] due to the presence of alkyl groups. In addition, with increment in AGO loadings, the thermal degradation temperature (TDT) for each composites displayed increasing pattern. For example, with only 1 wt% of AGO in AGO-10/PP, the TDT was increased more than 33°C when compared with PP. The reason for this increase pattern was due to the reduction in thermal contact resistance where it blocks the thermal conductivity at the interface between AGO and PP. AGO contained higher amount of alkyl chains, which showed strong interplanar interaction with PP and thus, decreasing the interplanar barrier between AGO and PP. Furthermore, good dispersion and high aspect ratio of AGO in PP increased the interaction site between AGO and PP.^[118] The same research group further investigated the thermal properties of alkylated carbon nanotube-reinforced polypropylene (ACNT/PP) composites and AGO-reinforced composites (AGO/PP) where the latter showed better thermal conductivity than the former. It was due to the presence of higher amount of alkyl groups on AGO than ACNT, which produced stronger interfacial adhesion than ACNT/PP in improving the thermal conductivity of AGO/PP composite.^[118] Furthermore, an increase in TDT due to increased amount of grafted GO with OTES also reported by Shin et al.^[126] This was attributed to GO contained large amount oxygen containing functional groups which enhanced the interfacial interaction between GO and PP matrix. Moreover, the organic functional groups of OTES formed the covalent bond with PP matrix. Therefore, the heat transfer was enhanced at the interface of GO and PP matrix which increased the TDT of composites. Besides PP-g-GO/MAPP nanocomposites

also showed the improvement in thermal properties as compared with MAPP. It was caused by all chains in PP-g-GO were entangled together and had similar molecular weights with MAPP.^[139] Thus, the addition of PP-g-GO improved the thermal stability of PP-g-GO/MAPP nanocomposites.^[140] Furthermore, the researchers also concluded that the well-dispersed GO act as a physical fence that limits the release of decomposed unstable products from MAPP matrix, improving the carbonization in the PP matrix.^[141]

3.4.4 | Fourier transform infrared

Fourier transform infrared (FTIR) is a useful characterization tool to recognize the presence of various functional groups of GO (Supplementary Figure 1). After modification with amine, two important characteristic peaks can be found at 2925 and 2855 cm⁻¹ which represented —CH asymmetric stretching and —CH symmetric stretching, respectively. These two peaks confirmed the presence of decylamines functional groups at the GO surface.^[94] Moreover, the peaks appeared at 1580 cm⁻¹ (C=O stretching of amide group), 1100 cm⁻¹ (C—N stretching) and 3500 cm⁻¹ (N—H stretching) implied the successful formation of amide linkages. Furthermore, the peak intensity at 2925 cm⁻¹ was increased when longer chain length of alkylamines was grafted at the GO plane.^[94] Besides amine group, GO can also be modified with alkyl group. AGO is one example where linear alkyl chains were introduced onto GO by S_N2 reaction between the oxygen groups at GO and the reactants. The appearance of alkyl groups at the GO surface was confirmed by the presence of CH₂ stretching vibration at 2924 and 2852 cm⁻¹, and the CH₃ degeneration at 1457 cm⁻¹.^[118]

A comparison of GO, GO-PPD, fGO, and MAPP-g-fGO was made by Yuan et al.^[108] After GO-PPD was reduced, most of characteristic peaks of GO were disappeared and two new alkyl chains peaks were detected at 1578 cm⁻¹ (N—H bending vibration) and 1162 cm⁻¹ (C—N stretching vibration) which confirmed the existence of the S_N2 reaction^[128] where the amine group was reacted with epoxide group of GO in S_N2 reaction.^[142] For fGO, another peaks can be observed at 1500 and 800 cm⁻¹ which attributed to the stretching of C=N and skeleton vibration of the triazine ring, respectively.^[108] After fGO was grafted with MAPP, the appearance of MAPP-g-fGO was confirmed by the strong and distinct peaks between 3002 and 2763 cm⁻¹ due to the presence of asymmetric and symmetric stretching vibration of methyl and methylene groups.^[143] Another characteristic peak was noticed at 1576 cm⁻¹ which attributed to secondary amide N—H bending vibration.^[144] The wide

absorption peak of N—H was overlapped with the secondary amide C=O stretching vibration at about 1640 cm^{-1} .^[144] These characteristic peaks demonstrated the successful of chemical covalent modification in the reaction between NH_2 functional groups of fGO and MA groups of MAPP.^[108]

Reactive compatibilization is an efficient method to improve the compatibility between nanofillers and non-polar polymer by forming new and stronger chemical bonds than van der Waals and hydrogen bonds to enhance interfacial interaction.^[145] This method was shown by Song and co-workers where hydrophilic bis (3-aminopropyl)-terminated poly(ethylene glycol) (NH_2 -PEG- NH_2) was grafted onto GO (GO-PEG).^[123] GO was first modified with NH_2 -PEG- NH_2 to produce GO-PEG. After modification, new peaks can be observed in GO-PEG spectrum at 2867 and 1109 cm^{-1} , which attributed to C—H and C—O bonds.^[123,146] For MAPP spectrum, two peaks can be seen at 1776 and 1715 cm^{-1} which represented C=O stretching of MA and carboxylic acid groups, respectively.^[147] In addition, by grafting MAPP onto GO-PEG to produce PP-g-GO, two peaks with reduced intensity at 1770 and 1721 cm^{-1} can be observed. Besides, 2.0 wt% PP-g-GO spectrum also showed another new broad peak at 3433 cm^{-1} , consistent with peak observed at 3429 cm^{-1} in GO-PEG spectrum. These observations proved the successful production of PP-g-GO.^[123]

The effect of different chain lengths of nanofiller, such as alkylamines, was reported.^[94] With addition 1 wt % of long chain alkylamine-modified GO, the intensity peak at 1650 cm^{-1} was increased with increasing chain lengths which also confirmed the successful attachment of amide functional groups. By using MA as compatibilizer, two peaks can be noticed at 1650 and 1600 cm^{-1} , which represented C—C stretching of MA and confirmed that maleic anhydride grafted PP (PP-g-MA) presented in samples.^[148] Furthermore, the addition of pristine GO has no effect on the changes of peak intensity from 1650 until 1600 cm^{-1} .^[94]

3.4.5 | Raman spectroscopy

The Raman spectroscopy is one of the important characterization tools to examine the molecular morphology of carbon materials, such as graphite, fullerene, CNTs, and graphene. Therefore, chemical modification which causes the changes of D and G peaks locations can be examined by Raman spectroscopy. For pristine GO spectrum, two peaks were appeared at 1352 and 1586 cm^{-1} , which corresponded to D and G peaks, respectively. However, for G peak, it was blueshifted to $\sim 6\text{ cm}^{-1}$ for both long chain

alkylamine-modified GO (GO-A10 and GO-A18), when compared with pristine GO as shown in Supplementary Figure 2. It is because the dopants attached onto graphene can influence two different conditions: G band position will shifts either to lower or to higher frequency when graphene behaves as electron donor or acceptor, respectively.^[149] Therefore, the alkylamines grafted graphene behaves as an electron donor due to the long chain alkylamines causes positive inductive reaction at the amine group grafted onto GO. Besides, no major different of I_D/I_G values for all three samples, concluding that long chain alkylamines were grafted either via epoxy-amine or amidation reactions that took places at the defect sites of GO.^[94] Moreover, by using Raman spectroscopy, the level of defect structure of sample can be investigated by calculating the I_D/I_G ratio of alkylated-grafted-GO (aGO) where it was increased as compared with GO after modified GO with octadecyl isocyanate ($\text{CH}_3(\text{CH}_2)_{17}\text{NCO}$) which proved that more amount of defects were created on aGO after alkylation.^[112]

Similar high values of I_D/I_G ratio were reported for GO-PEG and 2.0 wt% PP-g-GO.^[123] The I_D/I_G value for GO-PEG was calculated to be 0.98, higher than 0.72 of I_D/I_G value for GO. This large increment confirmed that attachment of NH_2 – PEG – NH_2 onto GO was occurred and created defects at the same time.^[150] For 2.0 wt% PP-g-GO, the I_D/I_G value was calculated to be 1.04 which increased from 0.98 (for GO-PEG). This was due to an increase of amount of defect structure during the grafting of MAPP onto GO-PEG in thermal reduction.^[34] Furthermore, G bands for both GO-PEG and 2.0 wt% PP-g-GO were downshifted by 9.0 and 12 cm^{-1} , respectively, as compared with GO. This trend occurred because of the elongation of carbon-carbon bonds during the tensile strain was conducted.^[151] The same elongation of bonds was happened to functionalization of NH_2 – PEG – NH_2 onto GO and attachment of MAPP onto GO. In addition, some reports suggested that I_D/I_G is inversely proportional with crystallite size (L_a) by using the equation: $L_a\text{ (nm)} = (2.4 \times 10^{-10}) \times \lambda_{\text{laser}}^4 (I_D/I_G)^{-1}$ where λ_{laser} is the laser excitation wavelength. This can be used to determine the amount of defect of GO based on the crystallize size.^[152,153]

3.4.6 | X-ray photoelectron spectroscopy

XPS is one of the popular characterization tools to investigate the relative quantities of carbon, oxygen, and functional groups in GO, RGO, AGO and ARGO. It is also an appropriate technique to examine the quantity of carbon and oxygen as compared to elemental analysis because the GO, RGO, AGO and ARGO are hardly to be fully

dehydrated.^[154] The XPS analysis showed that an increase in peak intensity of C—C bonds in AGO sample, proving alkyl groups were successfully grafted onto GO because alkyl groups have higher carbon percentage.^[118] The similar result also reported by Zhang et al^[155] where alkylated graphene oxide (GO-HD) displayed higher C/O percentage than GO which originated from the attached long alkyl chains.

Besides XRD and FTIR, the successful of covalent functionalization of $\text{NH}_2\text{-PEG-NH}_2$ onto GO and followed by reduction of GO during attachment of MAPP onto GO-PEG can be verified by using XPS where the presence of N 1 s peak and an increase in C 1 s peak were noticed in Supplementary Figure 3(A),(B). In Supplementary Figure 3(E), the atomic ratios of carbon and oxygen (C/O) were calculated to be 2.35:1 and 2.16:1 for GO and GO-PEG, respectively, indicating the $\text{NH}_2\text{-PEG-NH}_2$ was attached onto GO successfully. In addition, a blank GO-PEG (cGO-PEG) was prepared without the addition of MAPP and the C/O ratio was estimated as 2.38:1, which showed higher value than C/O ratios of both GO and GO-PEG. This result confirmed the GO was reduced during the grafting of MAPP onto GO-PEG and followed by solution blending with MAPP.^[123] Thus, C/O ratio is an important indicator to support both XRD and FTIR results.

3.4.7 | Mechanical properties

Homogenous dispersion and strong interlayer interactions between filler and matrix are important factors to enhance mechanical properties of the produced composites.^[156] PP is classified as a semi-crystalline polymer, which suitable for the synthesis of graphene-based PP composites with high Young's modulus and tensile strength values.^[54] Young's modulus can be defined as an indicator of plastic rigidity. Rigidity is influenced by the degree of crystallinity of a material. Therefore, a rigid material has excellent crystallinity and Young's modulus.^[157] Tensile strength is referred to the ability of a material to withstand an amount weight of loading.^[158] High value in tensile strength implies that the material can endure much more loading. Therefore, Young's modulus and tensile strength are very important factors especially in determining the right material for applications that requires rigid material and/or able to withstand high loading at the same moment. It was demonstrated that when the filler loadings and chain lengths of grafted alkylamines were increased, the Young's modulus and tensile strength of samples were increased as well.^[94] The degree of filler dispersion and interlayer interactions between filler and polymer were discussed as the reason

of improvement of mechanical properties.^[63] Besides, an increase in percentage crystallinity caused by nucleating efficiency of GO, dispersion degree of GO and interlayer interactions between PP matrix and GO were also explained as factors of improvement. However, when GO and modified GO were compared, the grafting of HMDA in GO showed better dispersion and interactions with PP matrix which responsible for the enhancement in mechanical properties.^[117]

On the contrary, the addition of AGO into PP matrix showed improvement in Young's modulus only but tensile strength and strain were reduced continuously with increasing AGO loadings. The strong interfacial interaction between the alkyl groups and the PP substrate as well as the presence of rough and wrinkled surface restricted the movement of the polymer chains near the AGO surfaces.^[118] In comparison between ACNT/PP composites and AGO-reinforced polypropylene (AGO/PP) composites, ACNT/PP showed increment of 20% and 37% in tensile strength and Young's modulus, respectively, with only 0.5 wt% of filler loading.^[118] The differences of shapes between AGO and ACNT were explained as the factors of enhancement in Young's modulus and decrement in tensile strength by AGO.

An interesting work was demonstrated by Dai et al^[113] where they demonstrated the formation of pores in composite during tensile test was conducted. The polyoxyethyleneoctylphenyl-10 (OP-10) was utilized as pore-forming agent to form pores in the stretched PP/GO/OP-10 composite membrane. A lot of primary pores were created with addition of OP-10 which improved the chain segment movement but inhibited the crystallization of PP. Furthermore, the initial pores facilitated the creation of new pores when stretched during the tensile test and produced bigger pores sizes for the stretched PP/OP-10 sample as compared with stretched pure PP. The highest percentage of porosity was obtained by stretching PP/GO/OP-10 with the addition of GO which worked as stress concentrator in helping the creation of more pores with smaller sizes.^[113]

Another research was reported by Dai and co-workers regarding the replacement of OP-10 with β -form nucleating agent (β -NA), that is, aryl amides compound (TMB-5).^[159] They found that β -NA showed higher nucleation efficiency than GO in the synthesis of β -PP. In addition, GO is the material causes the creation of pores at some conditions such as high stretching temperature and/or tensile speed. The similar observation also was highlighted by Shao and Mao in which the porosity of β -PP sample increased quickly during stretching in tensile test with a maximum at draw ratios around 2 and then, reduced afterward.^[160] The results demonstrated that the presence defect was caused by the formation of

pores in $\beta \sim$ crystalline polypropylenes under stretching condition. This phenomenon can be noticed in the pore formation in Celgard films and other polymers, such as the stress whitening and initiation of fractures due to those materials are defect induced. The abovementioned works demonstrated the synthesis of porous PP film where it could become the current research trend and opens up a new method to custom-make a product for some applications.

3.5 | Synthesis of ARGO and ARGO/PP composite

Alkylation of RGO is very similar with alkylation of GO where alkyl groups were introduced onto RGO planes to modify the chemical property for some applications.^[110] For example, TOAB and dodecyl iodide were used to graft alkylated groups onto GO and followed by reduction to synthesize ARGO.^[161] In another work, Qiu demonstrated the preparation of functionalized graphene (FG) using the 4,4'-diphenylmethane diisocyanate (MDI) to functionalize GO, before the functionalized GO was reduced using stearic acid, where the main purpose was to enhance the mixing compatibility with iPP.^[81] To produce FG/iPP composite, both FGs and iPP were melt-mixed in a Haaker mixer with five different FG loadings. However, isocyanates is not highly recommended to be used, as it is very toxic.^[161] Besides diisocyanate, a study was described by You and co-workers on the modification of RGO by using ethylenediamine. The produced alkylated RGO, denoted as rGO-NH₂, was synthesized prior to the reaction with MAPP to prepare polypropylene grafted reduced graphene oxide (PP-g-rGO). PP-g-rGO was produced by mixing amino groups of rGO-NH₂ and anhydrides of MAPP in an amidation reaction. PP-g-rGO also acts as a compatibilizer for blending between PP and polystyrene.^[162] On the other hand, most research works reported the functionalization of GO before the reduction is carried out. However, Bian and coworkers demonstrated opposite way in which the reduction of GO by water was conducted before the functionalization of rGO using octadecylamine (ODA) for the purpose to mix with PP in solution mixing method.^[163]

A new method combines both reduction and alkylation steps to synthesize alkylated and reduced graphene oxide nanoribbons (A-rGONRs), namely Dean-Stark trap method was reported by Song et al.^[164] This method incorporated four processes into a single step. The four processes were alkylation (octylamine was used), reduction, dispersion of graphene oxide nanoribbons (GONRs) in xylene and the mixing between iPP and A-rGONRs.^[164] GONRs were synthesized by opening

CNTs, followed by the alkylation, where almost 25 wt% alkyl chains were successfully grafted onto the surface of produced A-rGONRs. The A-rGONRs were dispersed in xylene before mixing with iPP. The produced iPP/A-rGONRs nanocomposites showed improvements in tensile properties, TDT and crystallization temperature (T_c).^[164]

Yang et al.^[59] reported a work where GO were grafted using alkyltrimethoxysilane and followed by reduction using hydrazine hydrate. Alkyltrimethoxysilane is one of the silane-coupling agents, which acted as compatibilizer to modify RGO to become more lipophilic. There are two ways to graft silane coupling agents onto GO. One way is to utilize the nucleophilic S_N2 dislocation in the functionalization of epoxide of GO surface with amine groups of silane.^[20] The second method is to conduct the dehydration process for the hydroxyl groups of both silane and epoxy groups of GO.^[165] These approach seems to be an innovative approach since the interfacial bond between polymer and graphene can be improved.^[59] Both alkyltrimethoxysilane-modified RGO and PP were dissolved in xylene, before the mixture was stirred for 8 hours to produce RGO/PP nanocomposites.^[59] The work demonstrated the benefits of choosing appropriate and several types of silane to study the effects of alkyl chain lengths on the organophilic properties of the modified graphene.

Another functionalization method is to functionalize GO sheets with hexylamine first to form hexylamine-graphene oxide (HA-GO) before the reduction of GO using hydrazine monohydrate solution.^[166] The work highlighted the use of this method as the perfect arrangement in order to form conductive and good mechanical properties alkylated graphene paper. Functionalization with hexylamine was to reserve high quantity of oxygen functional groups at the GO sheets. If no hexylamine was used, the alkylated graphene paper became less-ordered and inconsistent in conductivity. On the other hand, if the reduction was completed prior to alkylation, majority of reactive oxygen-containing functional groups from GO were removed. Consequently, the alkylation by hexylamine on GO surface cannot be proceeded.^[166]

3.6 | Characterization of ARGO and ARGO/PP composites

3.6.1 | Scanning electron microscopy

Neat PP shows different morphology with ARGO in nanocomposite due to the formation of visible aggregation and the amount was increased with increasing ARGO loadings. Having this agglomerates, mechanical properties of the composites also were influenced.^[59,119]

In addition, the dispersion of graphene (Gs), GOs and functionalized GO (FGs) in the iPP matrix were compared and investigated where different morphologies can be seen clearly in SEM micrographs (Supplementary Figure 4).^[77] Serious aggregations of Gs and GOs in the matrix were observed in Supplementary Figure 4(A)–(D), respectively. The aggregation of Gs in Gs/iPP was due to the exfoliation and intercalation that barely happened in normal melt mixing. For GOs/iPP, it is because the mixing of GOs and iPP were unsuccessful in melt mixing. However, uniform distribution of FGs was seen in FGs/iPP composites in Supplementary Figure 4(E), (F) where the FGs appeared as single platelet form at the surface composite, which indicated well dispersion due to their improved interfacial interaction after alkyl functionalization.^[81]

3.6.2 | X-ray diffraction

XRD has been used for phase identification to provide conclusive structural information. The successful covalent grafting in the reaction between anhydride groups of MAPP and amino groups of rGO-NH₂ was displayed in Supplementary Figure 5(a) as conducted by You et al.^[162] They observed that GO displayed a peak at $2\theta = 8.9^\circ$ with 0.99 nm as interlayer distance. However, after modification with ethylenediamine, a broad peak can be seen at $2\theta = 24.9^\circ$ for rGO-NH₂ and the interlayer distance was reduced from 0.99 to 0.34 nm, indicating a restoration of graphite structure in rGO-NH₂.^[162] For PP-g-rGO, it was produced after further modification of rGO-NH₂ with MAPP and several new peaks were noticed which represented the crystal phase of PP and successful modification of MAPP chains onto rGO-NH₂.^[167] Besides, when different types of alkyl chains were used, the presence of peak location could be changed. Supplementary Figure 5 (b) shows the XRD spectra for reduced graphene functionalized with octadecylamine (rG-ODA) and graphene (G-ODA) which exhibited a broad peak at $2\theta = 21^\circ$, a lower angle than rGO (24°), which caused by successful grafting of ODA onto rG and graphene. Furthermore, G-ODA samples exhibited higher peak intensity around 21° as compared with rG-ODA, which proved that more amount of ODA were grafted onto G-ODA. Interestingly, a new peak at $2\theta = 3.3^\circ$ can be seen for G-ODA as compared with rG-ODA due to the presence of ODA at interfacial graphite layers which increased the d-spacing.^[163] Investigation of composite by XRD is crucial after the fillers were mixed with neat PP in melt condition using a compounder. The XRD analysis able to verify the effect of nanofiller loadings as reported by Bian et al.^[163] All samples exhibited almost similar

characteristic peaks, except for the peak intensity at $2\theta = 21^\circ$, where the intensity of peak increased with increasing loadings of both G-ODA and rG-ODA which overlapped with the characteristic peak for neat PP (Supplementary Figure 5(C)).^[163]

3.6.3 | Thermogravimetric analysis

TGA provides detailed measurement of weight changes as a function of temperature or time in a measured environment. Supplementary Figure 6(A) shows the weight changes of graphite oxide, rGO-NH₂, and MAPP with temperatures. A weight loss of 19.9% was observed at 500°C for rGO-NH₂, much lower as compared to graphite oxide. This pattern proved that oxygen-containing functional groups in rGO-NH₂ were removed substantially.^[162] The thermal characteristic of GONRs and A-rGONRs are interesting because their shapes almost same with carbon nanotubes. In Supplementary Figure 6(B), GONRs experienced $\sim 28\%$ weight loss at temperature between 100 and 170°C which caused by the removal of oxygen containing functional groups. Furthermore, GONRs continued experiencing weight loss of ca. 18% between 170 and 270°C because of decomposition of defect carbon structures. On the other hand, from 210 to 290°C , A-rGONRs exhibited noticeable weight loss of ca. 25%. This loss might be due to pyrolysis of alkyl groups that grafted onto A-rGONRs. The weight loss resulted from thermal decomposition of carbon structures happened in much wider temperature range from 290 to 600°C , proving that A-rGONRs have better thermal stability than GONRs.^[164]

3.6.4 | Fourier transform infrared

The successful of alkylation onto GO or RGO sheets can be verified by using FTIR spectroscopy. For instance, Supplementary Figure 7(A) compared the FTIR spectra for GO, RGO, RGO-3C, RGO-8C, and RGO-12C. Propyltrimethoxysilane, octyltrimethoxysilane, and dodecyltrimethoxysilane were grafted onto the RGO sheets and they named as RGO-3C, RGO-8C, and RGO-12C, respectively. The addition of silane as compatibilizer to enhance the interfacial interaction between polymer and graphene by grafting the silane coupling agent onto graphene was demonstrated by Yang et al.^[59] Furthermore, the researchers also found that with an increase of the alkyl chains length, the peak intensity of C–H stretching was increased evenly.^[59]

If multi-walled carbon nanotubes (MWCNTs) were unzipped, the tube shape will be layered into plane sheets similar with graphene sheets. Song et al.^[164]

demonstrated their work in three steps: unzipping MWCNTs, alkylation, and reduction of samples. Both alkylation and reduction were conducted by using a Dean-Stark trap. Supplementary Figure 7(B) shows FTIR spectra for GONRs and A-rGONRs. After reduction, the intensity for hydroxyl functional group at wavelength 3410 cm^{-1} decreased, and carboxyl functional group was no longer presented in A-rGONRs. However, the peaks intensities corresponded to C—H bond were increased. It is because octylamine attacked the epoxide groups of GONRs, which caused a nucleophilic reaction where the alkyl groups were grafted onto GONRs and at the same time eliminated oxygen groups.^[164] In addition, nucleophilic reaction also mentioned by Bian et al.^[163] to elucidate the graft mechanism between ODA and epoxy groups. In Supplementary Figure 7(C), the FTIR spectra of GO, ODA, rG-ODA, and rGO were compared. By refluxing the samples at high temperature condition, carboxyl groups of GO were decomposed into CO_2 by hydroxyl (—OH) of water which was also reported by Fan et al.^[168] Besides, in this reaction condition, carboxyl groups were removed and the epoxy groups were remained. Therefore, the ODA was grafted successfully onto GO with the presence of epoxy via nucleophilic ring-opening reaction.^[163] Recent study by Eskandari et al delineates the polymer grafting strategy carried out on the graphene layers added further insight to this area of study.^[169]

3.6.5 | Raman spectroscopy

Raman spectroscopy is an important tool to investigate the presence of defect of carbon structure in sample because the alkylation induces more formation of defects on GONRs and rGONRs sheets. The spectra contain sharp D and G peaks at 1350 and 1580 cm^{-1} , respectively (Supplementary Figure 7(D)). Higher intensity of G and D peaks can be seen for GONRs as compared with A-rGONRs. Furthermore, the Raman ratio values (I_G/I_D) were calculated as 0.82 and 1.10 for GONRs and ArGONRs, respectively. Therefore, A-rGONRs had better quality of carbon structure than GONRs.^[164]

3.6.6 | X-ray photoelectron spectroscopy

An investigation about the successful of alkylation and reduction is important as shown in Supplementary Figure 8. In Supplementary Figure 8(A),(B), three peaks can be seen at 284.4 , 286.6 , and 288.5 eV ^[164] which corresponded to C—C bonding, C—O bonding, and C=O bonding, respectively. However, the peak intensities for

C—O and C=O bondings were decreased in reduced GONRs (A-rGONRs) spectrum as compared with oxygen functionalized GONRs spectrum. This decrement pattern can be witnessed in the O 1 s spectra as well, as displayed in Supplementary Figure 8(C),(D). The comparative intensity ratio of C—O to C=O bondings was decreased, thus proving that the reduction of GONRs, especially C—O bonding was due to octylamine reacted with epoxide groups from GONRs during nucleophilic attachment of alkyl groups onto GONRs and at the same time, oxygen groups were removed. Reduction was initiated at high temperature (130°C) where major oxygen content loss was noticed from 29.1% in GONRs to 9.6% in A-rGONRs. In Supplementary Figure 8(e), XPS N 1 s spectrum of A-rGONRs has two peaks, which attributed to C—N bonding and N—C=O bonding, at 399.6 and 400.1 eV , respectively. The amount of nitrogen contained in A-rGONRs was increased as compared with GONRs, confirming that octylamine was grafted onto GONRs surface.^[164]

3.6.7 | Mechanical properties

To produce composite with superior mechanical properties, traditional enhancement by adding nanofiller into neat polymer is a common approach. However, to enhance the dispersion of nanofillers in matrix due to different polarity, alkylation of nanofiller is a crucial step. The presence of oxygen containing functional groups of GO enhances the interactions between GO and polar polymer. In contrast, GO needs surface modification by grafting alkyl chains onto the GO surface to achieve the good interaction with non-polar polymers. Thus, the improved mechanical properties have been studied and reported recently by some researchers. Song et al.^[164] reported that A-rGONRs showed better reinforcement in the Young's modulus, tensile strength and elongation at break as compared with rGONRs for the same loading. It was due to A-rGONRs have bigger interfacial sites, higher aspect ratios and more abundant amount of alkyl groups than rGONRs that contributed to the good interfacial interaction with neat iPP matrix.^[164] In addition, the uniform and random distribution of A-rGONRs in matrix were another factors facilitated the load transfer from iPP to the filler effectively. They also reported that increment of nanofiller loadings enhanced the mechanical properties of produced nanocomposites significantly.^[164]

In another example, FGs were prepared from the reduction of diphenylmethane MDI-GO using stearic acid before mixing with iPP to prepare iPP/FGs composite.^[81] The addition of graphite flakes (Gs) into iPP matrix resulted increment in the Young's modulus and yield

strength values. This was due to an increase in crystallinity of iPP, implying that the Gs behave as source to activate the crystallization of iPP. However, with the formation of aggregates and weak interplanar interaction between iPP and Gs, the elongation at break values decreased. This decrement trend continued as the content of Gs increased. Thus, iPP/Gs lost its toughness and showed fragile structure.^[81] When GOs were used as filler, the value for Young's modulus was increased. Nevertheless, the improvement in yield strength for iPP/GOs composites was not as significant as compared with iPP/Gs because of low crystallinity in iPP/GOs. In addition, the decrement pattern in elongation at break as noticed in iPP/Gs can be noticed in iPP/GOs as well. However, the values were improved significantly in iPP/GOs than that of iPP/Gs for all fillers loadings. For example, for 0.1 wt% filler loading, the elongation at break of iPP/GOs was 15 times superior than iPP/Gs. This caused by low crystallinity and wrinkles shape of GOs, which led to better interfacial interaction between iPP and GOs.^[81] The best mechanical properties were achieved by iPP/FGs, where it exhibited high Young's modulus and yield strength. Its elongation at break was relatively high, compared to iPP/Gs and iPP/GOs, implying that FGs were dispersed well in iPP matrix and interacted strongly with iPP.^[81]

Different approaches to mix single type of alkylated nanofiller with single type of polymer in compatible manner were reported mostly. However, mixing of two types of polymers could be a challenge to enhance the compability between nanofilers and polymers. A work regarding the mixing of PP and polystyrene by using compatibilizer of polypropylene-graft-reduced graphene oxide (PP-g-rGO) was demonstrated by You and co-workers. The enhancement in both tensile strength and elongation at break as shown in Supplementary Figure 9 (A) was noticed clearly with the increase of PP-g-rGO loadings. Without PP-g-rGO, the value of tensile strength for PP/PS blend was even lower than neat PP while the value of elongation at break was also very low. This indicated the addition of PP-g-rGO improved the interfacial interaction between PP-g-rGO and PP/PS. Besides, the presence of PP-g-rGO reduced the sizes of PS particles, which induced the interplanar stress transfer. Furthermore, PP-g-rGO increased the stiffness of PP/PS composites due to high Young's modulus of graphene as proved by an increase in storage modulus of PP/PS/PG nanocomposites with the increasing content of PP-g-rGO.^[162]

Besides, long alkyl chain such as propyltrimethoxysilane was grafted onto RGO and named as RGO-12C to improve the solubility of RGO in organic solvent successfully. However, the improvement of long alkyl chain on the mechanical properties could puzzled researchers and

finally, the answer was reported by Yang et al.^[59] (Supplementary Figure 9(B)). The reinforcement in both tensile strength and Young's modulus increased with increasing of RGO-12C loadings. In contrast, elongation at break increased at the beginning, but decreased as the RGO-12C loadings were increased. The enhancement in mechanical properties of composites was attributed to the presence of unique mechanical properties of RGO-12C. In addition, homogenous dispersion of RGO-12C in PP matrix enhanced the interplanar interaction to enable the stress transfer from PP matrix to RGO-12C.^[73] When the amount RGO-12C was increased to maximum loading at 3 wt%, opposite trends of mechanical properties were observed due to the agglomeration of RGO-12C which reduced the interaction between nanofiller and PP matrix and further formed defects structures at the contact area between RGO-12C and PP matrix.^[59]

4 | APPLICATION OF GRAPHENE DERIVATIVES POLYPROPYLENE NANOCOMPOSITES

4.1 | Electronic devices

Graphene electrodes have been widely applied in various industrial applications at acceptable cost.^[170,171] GO shows good electrical conductivity after it is being reduced by thermal or chemical reduction, or combination of both reduction methods.^[172–178] The successful of restoration of electrical conductivity significantly depends on the efficiency of elimination of oxygen containing functional groups from the edge and basal plane of GO.^[172] Thus, GO based polymer composites are potential material for the application in electronic devices if the restoration of graphene electrical property is achieved. GO based PP nanocomposites were synthesized by in situ polymerization where a supported catalyst was used to induce the mixing process between non-polar PP and polar GO.^[67] Even though the electrical result displayed low conductivity of 0.3 S m^{-1} at a 4.9 wt% loading, the utilization of supported catalyst enables the mixing between GO with different polar of polymers.^[179]

4.2 | Electromagnetic interference shielding

Electromagnetic interference shielding utilizes conducting polymer composites as new potential material to shield electronic devices from electromagnetic interference/radio frequency interference, which is released by high voltage power cables, laptop, desktop computer,

cellular phones, radio transmitters, and others. It is because conducting polymer composites have some better properties such as anti-corrosion, low cost, light-weight, low density, electrical conductivity, and elasticity than those existing metal, which used as electromagnetic interference shield.^[180] The corrosion, low rigidity, and heavy weight of metal are those disadvantage factors which could not be found in conducting polymer composites. There are reports highlighted that if resistivity of a composite lower than $10 \Omega \text{ sq}^{-1}$, it has potential to be used as an effective electromagnetic interference shield.^[181,182] It is because the addition of conducting nanofillers can improve the electrical conductivity by forming conductive networks to a much higher value than its electrical percolation threshold.^[183–186] The properties of PP have been improved by adding several types of nanofibers and nanofillers for the past few years to synthesize composites with enhanced properties.^[74] With the discovery of graphene, possible applications of graphene-based PP composites in various research fields have been widely reported due to the improved thermal, mechanical and electrical properties of composites. Therefore, the application has been expanded to electromagnetic interference shielding by mixing graphene with PP matrix.^[187,188] Panwar et al. demonstrated the synthesis of PP-graphite composites for EMI shielding material in which PP was used as polymer and graphite was used as conducting nanofiller. The shielding effectiveness value was measured as 44.12 dB at 2.76 GHz for composite with $\phi = 0.075$. Besides, the authors also highlighted that PP-graphite composite also suitable for charge storing devices and decoupling capacitors with its highest dielectric constant of 4.04×10^5 and dissipation factor of 7.53×10^4 , respectively which measured at 40 Hz for composite with $\phi = 0.075$.^[180]

4.3 | Flame retardancy

Various types of nanofillers have been incorporated with polymer to form composites for wide applications which used by peoples daily. It is because polymer composite shows low ratio of weight to strength, anti-corrosion, anti-chemical, heat insulator, good physical, and other excellent properties.^[189,190] However, polymer composites are naturally flammable when they are burned and release poisonous gases which harmful to human health.^[191–193] Thus, it is crucial to enhance the flame retardancy of polymers composites to fulfill safety usage guidelines. Several strategies to improve flame retardancy have been reported such as coating of fire-retardant layer, addition of flame-retardant filler and immersion in flame-retardant solution.^[191,194,195] Among these

approaches, the addition of flame-retardant filler such as graphene and its derivatives show promising effectiveness in reducing the flammability of polymer composites by enhancing thermal stability of polymer composite, slow down its burning, prohibit fire from propagating and decrease heat release rate.^[196,197] Besides, they have good barrier effect and huge specific surface adsorption ability that helping the decrease in heat and mass transfer.^[51] Furthermore, Yuan et al. showed the effectiveness of application of RGO at low loadings of 0.5 to 1 wt% in RGO/iPP composites to reduce the heat and smoke release because of the improvement of swelling of char which cause the superior in isolation of char. Furthermore, the compactness of char for RGO/iPP composites was improved based on the observation from morphology of char residues.^[198]

5 | CONCLUSION

There are numerous approaches to prepare graphene-based polypropylene nanocomposites by employing the advantages of unique properties of functionalized graphene. However, the most serious issues are the poor dispersion, which causes the weak interlayer interaction of graphene with PP matrix and thus, greatly reduces the mechanical properties of composites. Regardless of various methods have been demonstrated such as melt processing, in situ polymerization, solution processing, and chemical functionalization, the improvement of nanofiller dispersion and enhancement of interlayers interaction between nanofiller and PP matrix are still available to be advanced in the future. Therefore, the functionalization of graphene to achieve the good dispersion and strong interfacial interaction between graphene and PP matrix are most important challenges. The successful improvement of interlayers adhesion between graphene and PP matrix, and at the same time enhance the dispersion of graphene in the PP matrix have been proved through the functionalization of graphene, such as alkylation, using different types of alkyl chains and lengths. Another encounter issue is the mechanical properties of graphene-PP composites could be affected by the presence of defective sites at functionalized graphene, which enhance or reduce the graphene-PP's interaction and nanofillers dispersion. Thus, choosing of the right alkyl chains and lengths for chemical alkylation of graphene are necessary to counteract mixing compability issue. As demonstrated by Ryu and Shanmugaraj,^[94] Young's modulus and tensile strength were improved with increasing the wt% of pristine graphene oxide (GO) or long chain, alkylamine-modified GO. For various applications, the use of graphene or GO or RGO or

TABLE 3 Comparison of graphene derivatives from their chemicals used, different ways of preparation methods, advantages and disadvantages

Graphene derivatives	GO	RGO	AGO	ARGO
Chemicals used	H_2SO_4 , H_3PO_4 , $KMnO_4$ ^[22–24]	Hydrogen sulphide (H_2S), ^[31] hydrazine and hydroquinone, ^[32,33] Hydrazine, ^[26] sodium bisulfite ($NaHSO_3$), sodium sulfite (Na_2SO_3), sodium thiosulfate ($Na_2S_2O_3$), sodium sulfide nonahydrate ($Na_2S \cdot 9H_2O$), thionyl chloride ($SOCl_2$), and sulphur dioxide (SO_2), ^[36] hydroquinone, ^[32] sodium borohydride ($NaBH_4$), ^[33] mushroom extracts, ^[37] green tea ^[38] and carrot juice ^[39] , microorganisms (like <i>Escherichia coli</i>), ^[40] sugars (such as fructose, glucose and sucrose ^[41]), ascorbic acid, ^[42] lemon juice and vinegar, ^[43] phosphorous acid ^[44]	Aminoethyl-aminopropyl-trimethoxysilane, ^[111] octadecyl isocyanate ($CH_3(CH_2)_{17}NCO$), ^[112] polyoxyethyleneoctylphenyl-10 (OP-10), ^[113] hexylamine (A-6), decylamine (A-10), hexadecylamine (A-16) and octadecylamine (A-18), ^[94,116] hexamethylene diamine (HMDA), ^[117] p-phenylenediamine (PPD) and cyanuric chloride (CC), ^[108] tetra-n-octylammonium bromide (TOAB) and dodecyl iodide, ^[118,119] 2-(2-[dimethylamine]-ethoxy) ethanol (DMAE), ^[120] hydrophilic bis(3-aminopropyl)-terminated poly(ethylene glycol) (NH_2 -PEG- NH_2), ^[123] octyltriethoxysilane (OTES), ^[126]	Tetra-n-octylammonium bromide (TOAB) and dodecyl iodide, ^[161] 4,4'-diphenylmethane diisocyanate (MDI), ^[81] ethylenediamine, ^[162] octadecylamine (ODA), ^[163] octylamine, ^[164] alkyltrimethoxysilane, ^[59] hexylamine ^[166]
Methods	For improved Hummers' method, H_2SO_4/H_3PO_4 mixture at 9:1 ratio and $KMnO_4$ are required to perform reaction at room temperature.	Various inorganic and organic reducing agents have been used to reduce GO to RGO by refluxing the mixture for few hours in silicone oil bath. The longer reaction time, the higher amount of RGO can be obtained but the synthesis cost is increased.	Most methods involve the mixture of GO and chemicals which stirred in silicone oil bath according to required temperature and duration time.	The synthesized GO is reduced by using chemical reducing agents by stirring the mixture for several hours in silicone oil bath.
Advantages	The presence of oxygen containing functional groups are available for further functionalization using alkyl chains. Melt mixing and solvent mixing are suitable methods used for mixing between GO and polar polymer to prepare GO based polymer composites for improvement in mechanical property.	The removal of oxygen containing functional groups can recover or repair the conjugated graphitic network which shows improvement in electrical conductivity than GO. Besides, RGO has almost similar mechanical and optoelectronic properties to pristine graphene. The mixing of RGO and non-polar polymers can be conducted.	Strong interfacial adhesion to a polymer matrix can be achieved, therefore leading to the homogeneous dispersion of alkylated GO in a hydrophilic polymer matrix and efficient load transfer.	Alkylated RGO can be effective as a nanofiller for reinforcement in hydrophobic polymer. Homogeneous dispersion and efficient load transfer can be accomplished.

(Continues)

TABLE 3 (Continued)

Graphene derivatives	GO	RGO	AGO	ARGO
Disadvantages	Compatible with polar polymers only. Poor electrical conductivity.	Compatible with non-polar polymers only	Compatible with polar polymers only. The synthesis cost could be increased due to additional experimental works to graft the alkyl chains on the GO are required. The electrical conductivity is reduced due to the conversion of sp^2 carbon to sp^3 carbon structure.	Compatible with hydrophobic polymers only. Since the synthesis requires the reduction of alkylated GO, the synthesis cost is increased. The electrical conductivity of alkylated RGO also decreased because of the presence of sp^3 carbon structure.

Abbreviations: AGO, alkylated GO; ARGO, alkylated RGO; GO, graphene oxide; RGO, reduced graphene oxide.

alkylated (AGO) or ARGO is really influenced by the production cost due to the functionalization requires time and some chemicals to be completed. Moreover, to prepare the excellent mechanical performances of graphene-PP composites for particular application, a suitable graphene functionalization method for mixing with PP is crucial. In other words, the type of application and production cost are important factors to be considered. Table 3 shows the comparison of graphene derivatives from perspectives of chemical used, synthesis methods, benefits and deficiencies. This table would allow readers to get useful information quickly. Therefore, we hope that this review will inspire breakthrough development in the functionalization of graphene to synthesize graphene-based polypropylene nanocomposites with tailored mechanical properties for various applications.

6 | OUTLOOK

6.1 | Inhomogeneous dispersion and distribution of nanofillers

A proper homogenous distribution and dispersion of nanofillers in polymer matrix is the main concern for the impressive reinforcement of the mechanical, thermal, and electrical properties of composites. It is because the interfacial interaction between nanofillers and polymer could be optimized by the well distributed network and minimum nanofillers aggregation and thus, it represents a great challenge to all researchers. There are some published reports highlight the incompatible mixing of different polarities of nanofillers with polymer occurs in conventional approaches, such as solution and melt stirring-mixing which affects the distribution and dispersion of nanofillers. Some mixing methods, such as long hours and strong sonication, fast stirring and forced convection flow might solve the aggregation issue but it also damages to the soft and flexible large graphene derivatives sheets into smaller sizes, which significantly form negative impact to the improvement of mechanical, thermal and electrical properties of composites.^[199]

To solve the issues of inhomogeneous distribution and dispersion of graphene derivatives by traditional mixing techniques, layer by layer assembly of polymer and graphene components is one of the proposed methods which conducted by alternate adsorption of complementary parts from graphene derivatives and polymer suspension. This technique improves the dispersion and distribution of components in a layered structure and therefore, mechanical properties and interfacial interactions are enhanced. However, layered and thin materials in micrometer range can be formed preferable rather than

bulk and thick structure. In addition, in situ polymerization method also has been used which involve the mixing of graphene derivatives and monomers that can boost their capable exfoliation into mono layers in every place of polymer matrix induced by catalysis reactions. Besides, in situ polymerization facilitates the intercalation of graphene derivatives in various types of polymers, such as polyimides (PI), PS, PET, and PMMA, due to the polymerization of homogenous distributed monomers can be achieved at most places in polymer matrix. However, the graphene derivatives might not compatible when react in acidic solution and be reduced in extreme temperature or strong basic solution. Thus, some challenges and limitations occur in the in situ polymerization.

6.2 | Interfacial interactions between graphene derivatives and polymer matrix

Van der Waals forces, π - π interactions, hydrophobic and hydrophilic interactions are common bindings happen between the graphene derivatives and polymer matrix. However, these bindings are usually too weak to form a strong interface between graphene derivatives and polymer matrix.^[200,201] Thus, graphene has been functionalized into GO layers and other functionalized forms to enhance the interplanar interaction with numerous polymers for reinforcement of mechanical property. GO has many benefits than that of graphite and graphene for the incorporation of GO into nanocomposite, including facile and low-cost mixing process with polar polymer, and the capability for additional functionalization. It is because GO is grafted with rich oxygen-containing groups. In addition, the selection of a suitable polymer is crucial for mixing compatibility with functionalized graphene derivatives. However, chemical modification of GO with proper functionalized groups compatible with a polymer is an alternative for improved reinforcement of mechanical property. For heterogenous GO, it is utmost to be grafted with heterogeneous linker for binding with different types of functionalized groups to maximize the interplanar interaction.

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CONFLICT OF INTERESTS

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